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**Hsieh et al.**

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(54) **PIXEL ARRAY INCLUDING DARK PIXEL SENSORS**

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**H04N 25/633** (2023.01)  
**H10F 39/00** (2025.01)

(52) **U.S. Cl.**  
CPC ..... **H04N 17/002** (2013.01); **H04N 25/633** (2023.01); **H10F 39/024** (2025.01); **H10F 39/8057** (2025.01); **H10F 39/8063** (2025.01)

(58) **Field of Classification Search**

CPC .... H04N 25/633; H04N 17/002; H04N 25/63; H04N 25/76; H01L 27/14623; H01L 27/14685; H01L 27/14627; H01L 27/14603; H01L 27/1464; H01L 27/14645; H01L 27/14689

See application file for complete search history.

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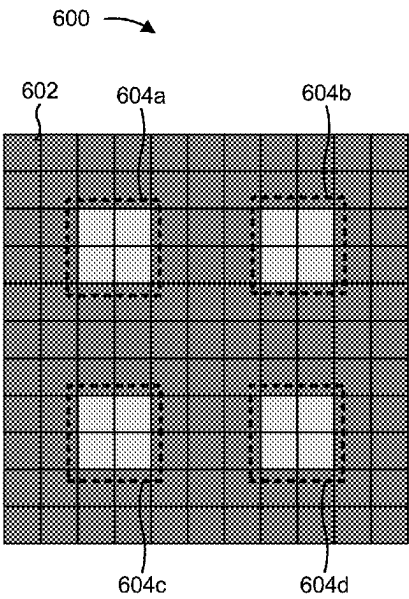
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(57) **ABSTRACT**

A pixel array includes a plurality of dark pixel sensors configured to generate dark current calibration information for a plurality of visible light pixel sensors included in the pixel array. The plurality of dark pixel sensors may generate respective dark current measurements for each of the plurality of visible light pixel sensors or for small subsets of the plurality of visible light pixel sensors. In this way, each of the plurality of visible light pixel sensors may be individually calibrated (or small subsets of the plurality of visible light pixel sensors may be individually calibrated) based on an estimated dark current experienced by each of the plurality of visible light pixel sensors. This may enable more accurate dark current calibration of the visible light pixel sensors included in the pixel array, and may be used to account for large differences in estimated dark currents for the visible light pixel sensors.

**20 Claims, 28 Drawing Sheets**



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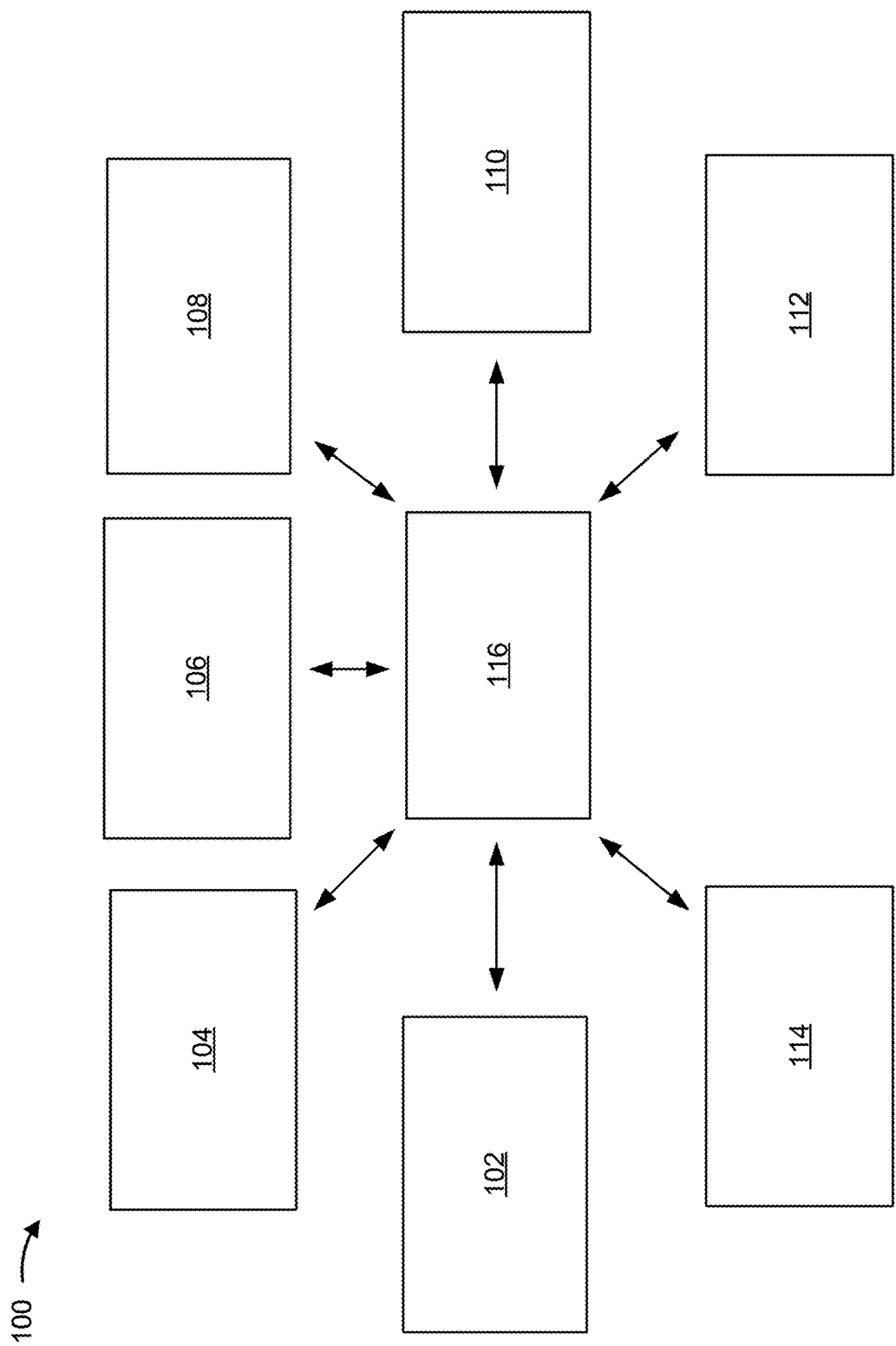


FIG. 1

200

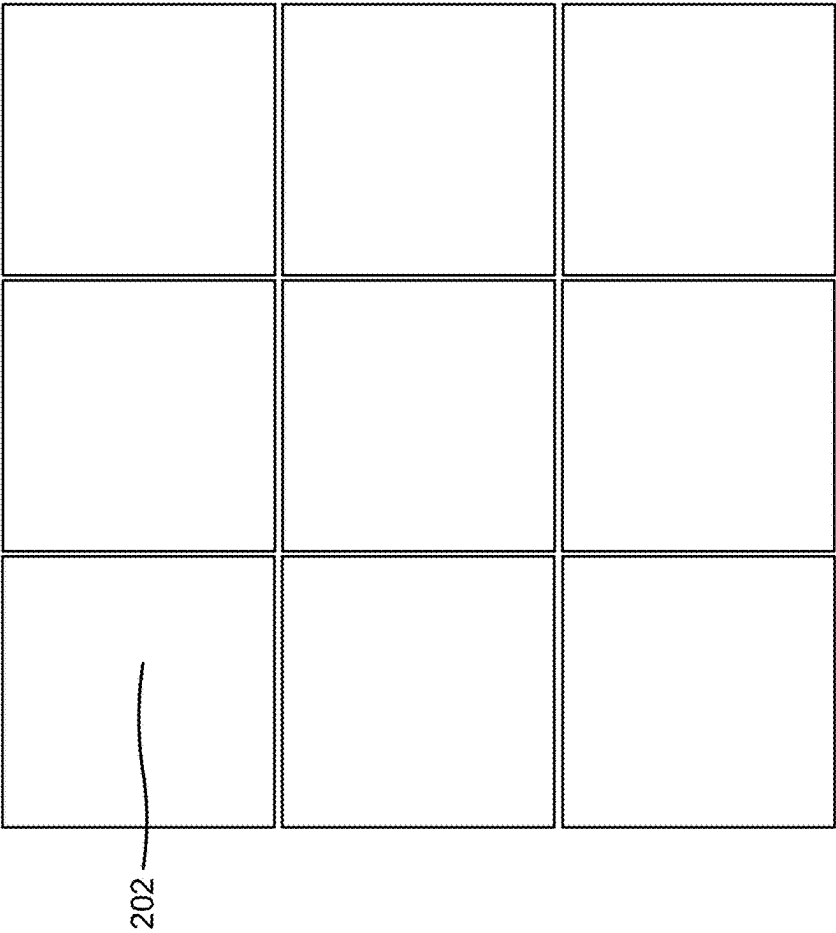


FIG. 2

300 →

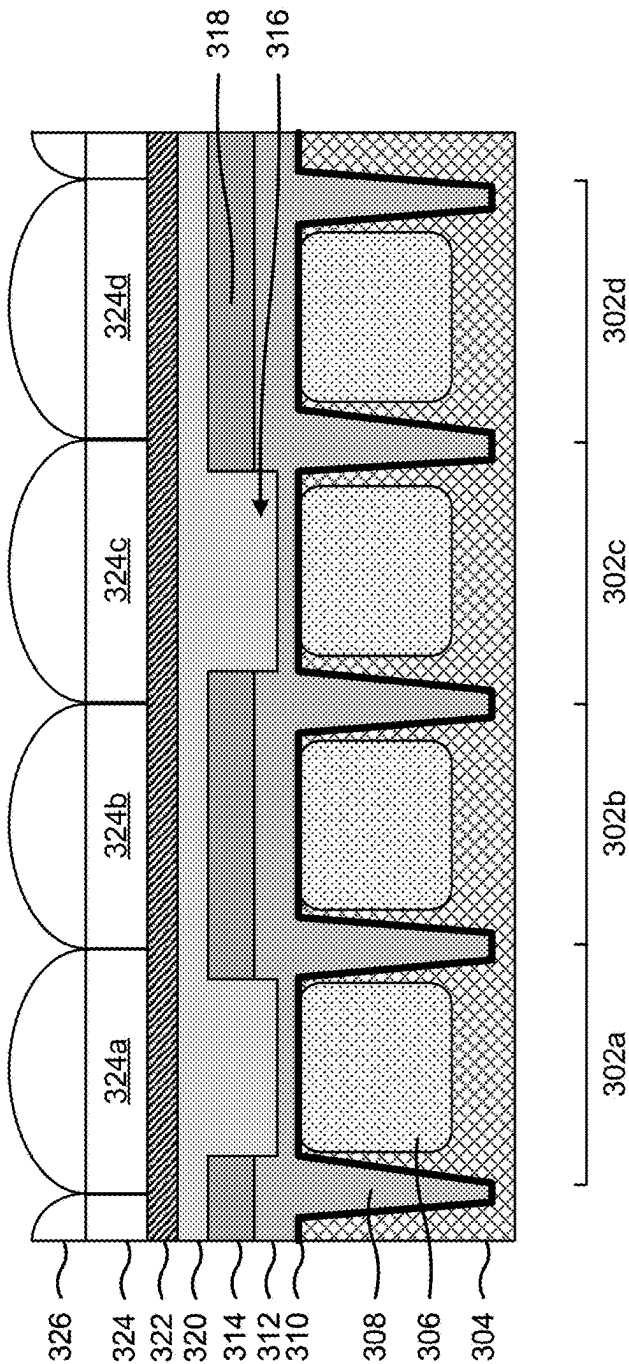


FIG. 3A

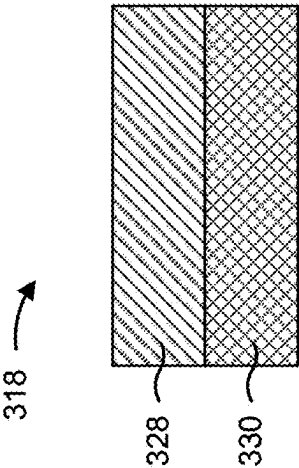


FIG. 3B

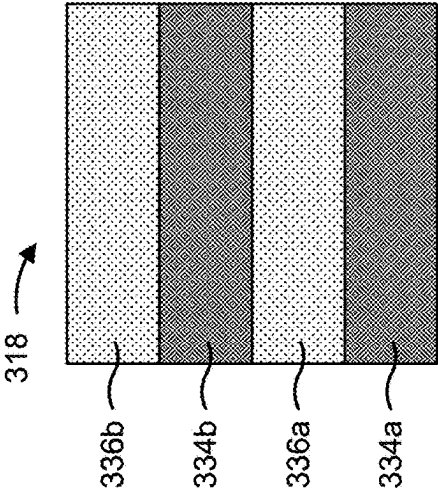


FIG. 3C

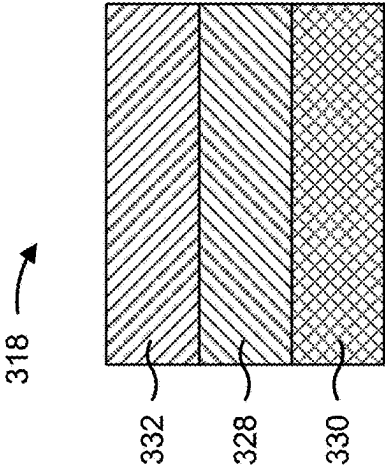


FIG. 3D

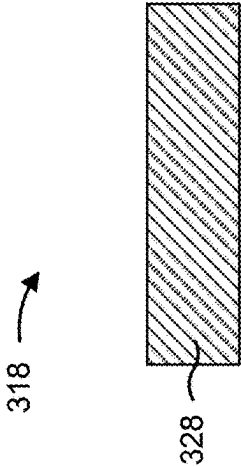


FIG. 3E

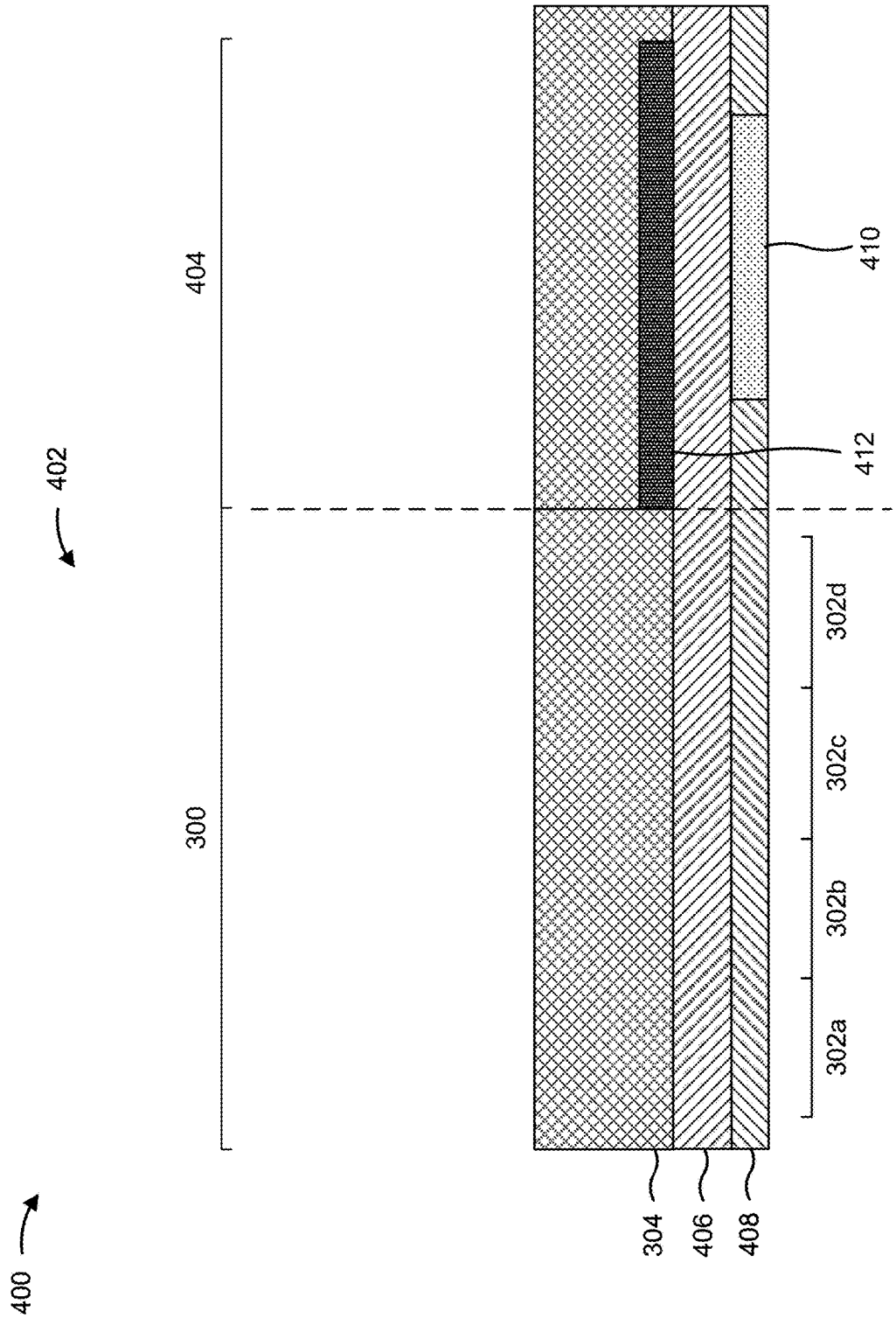


FIG. 4A

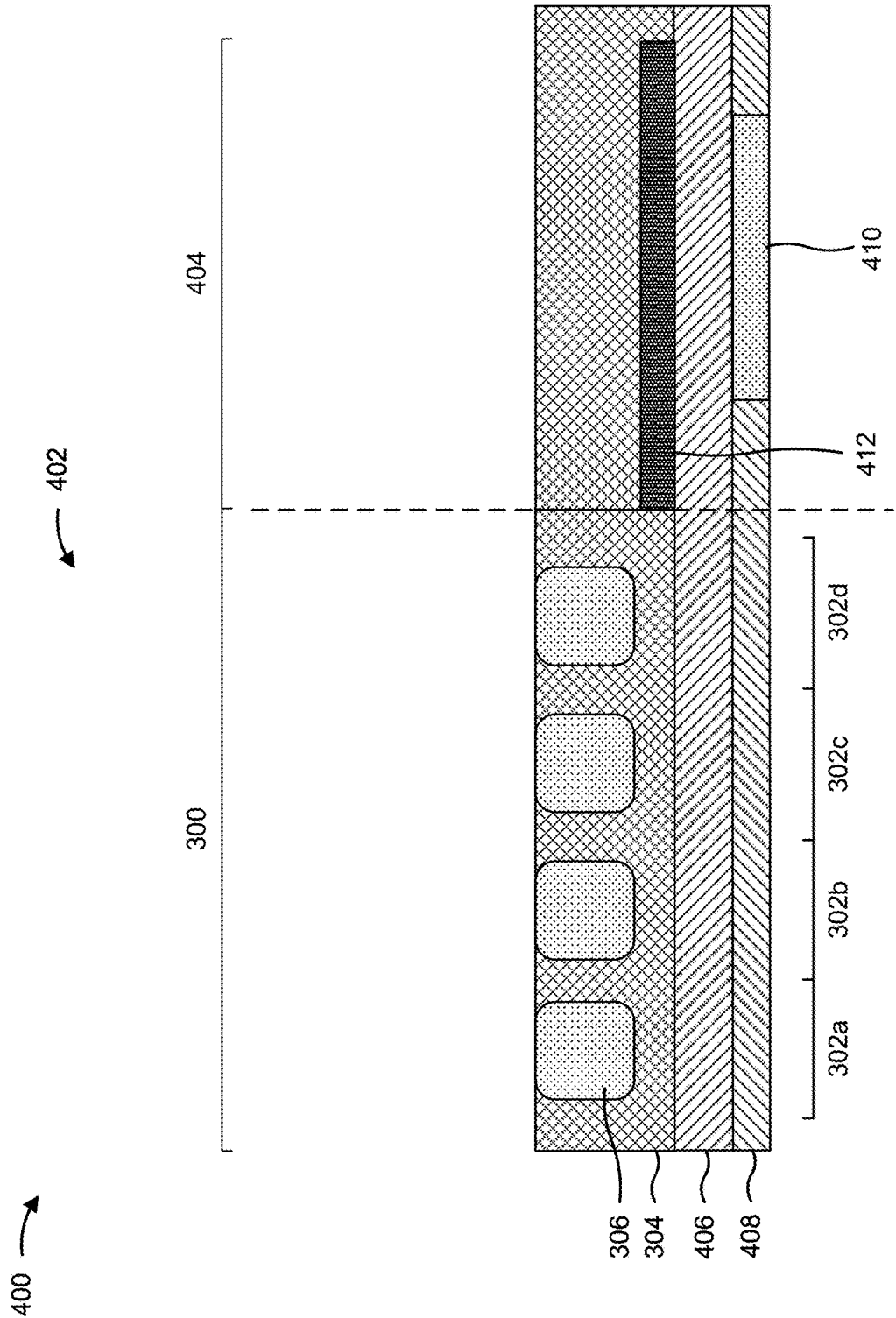


FIG. 4B



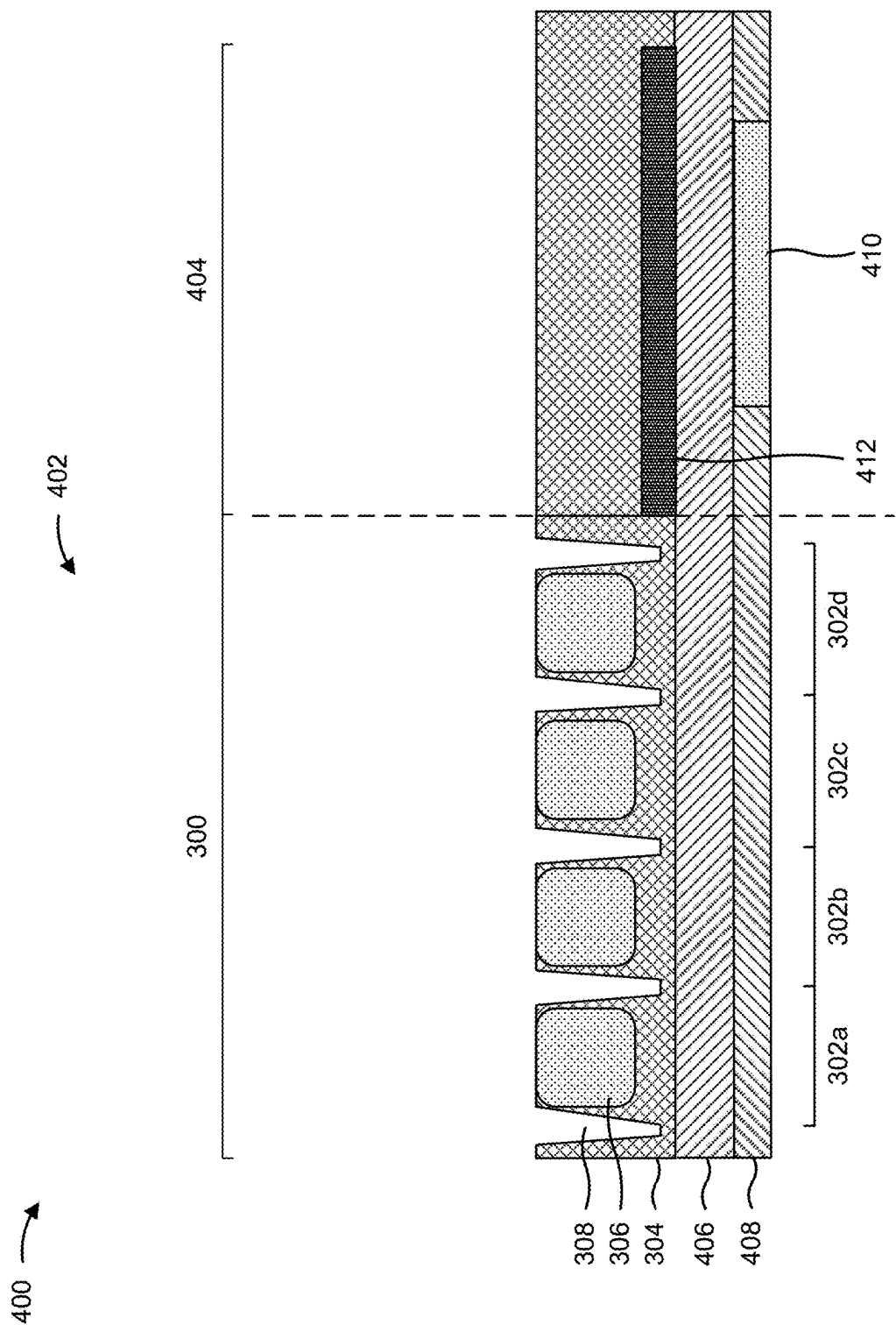


FIG. 4C

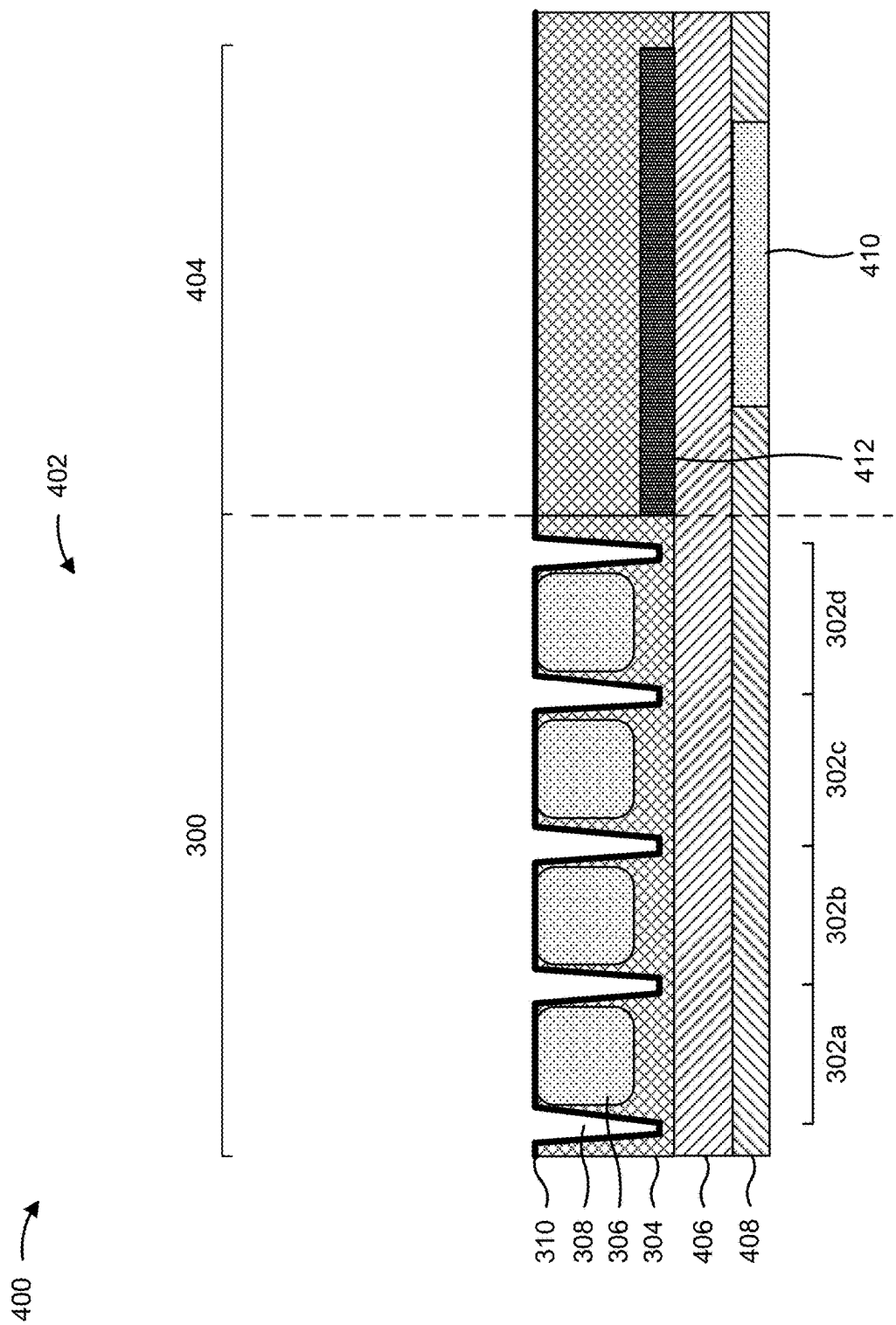


FIG. 4D

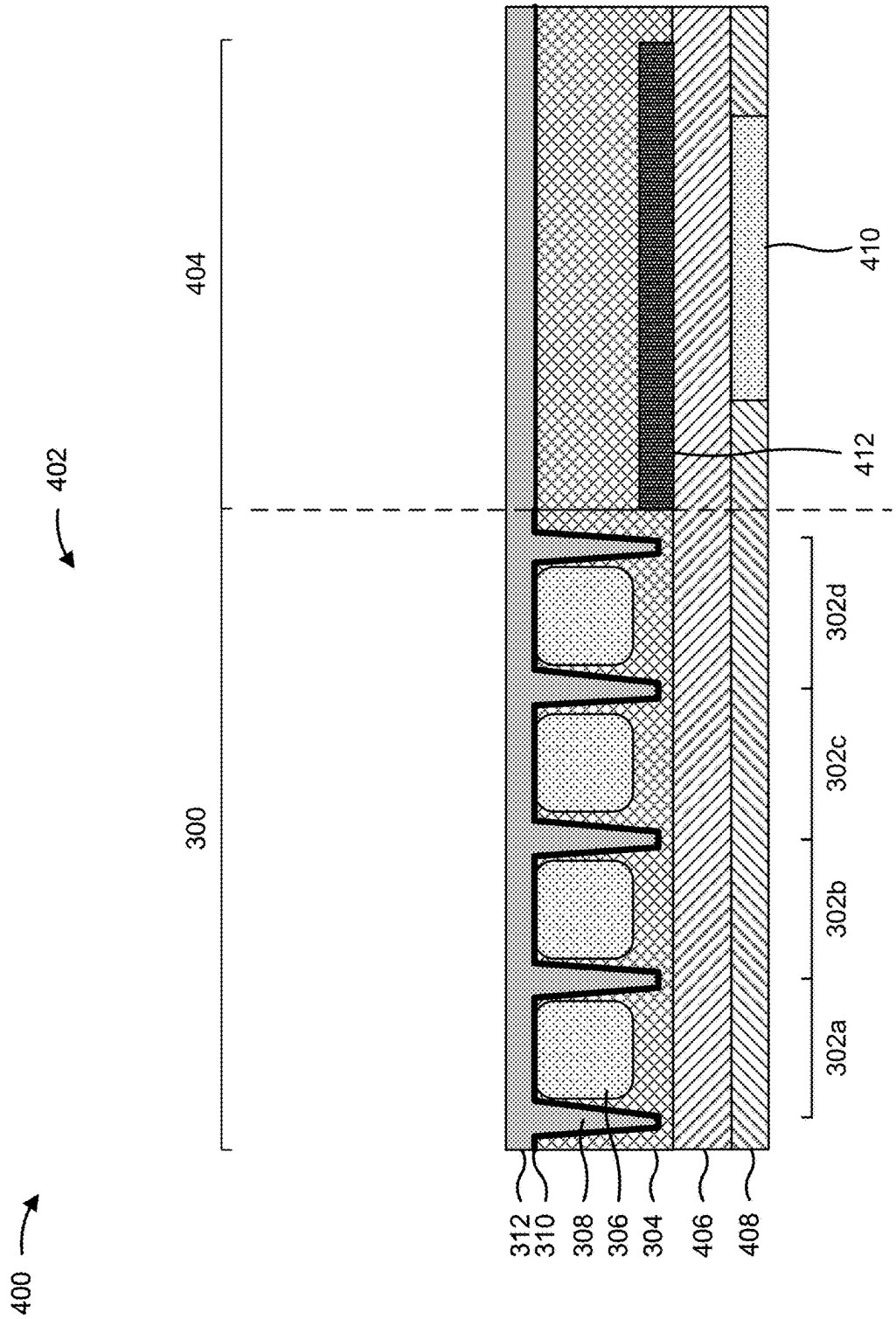


FIG. 4E

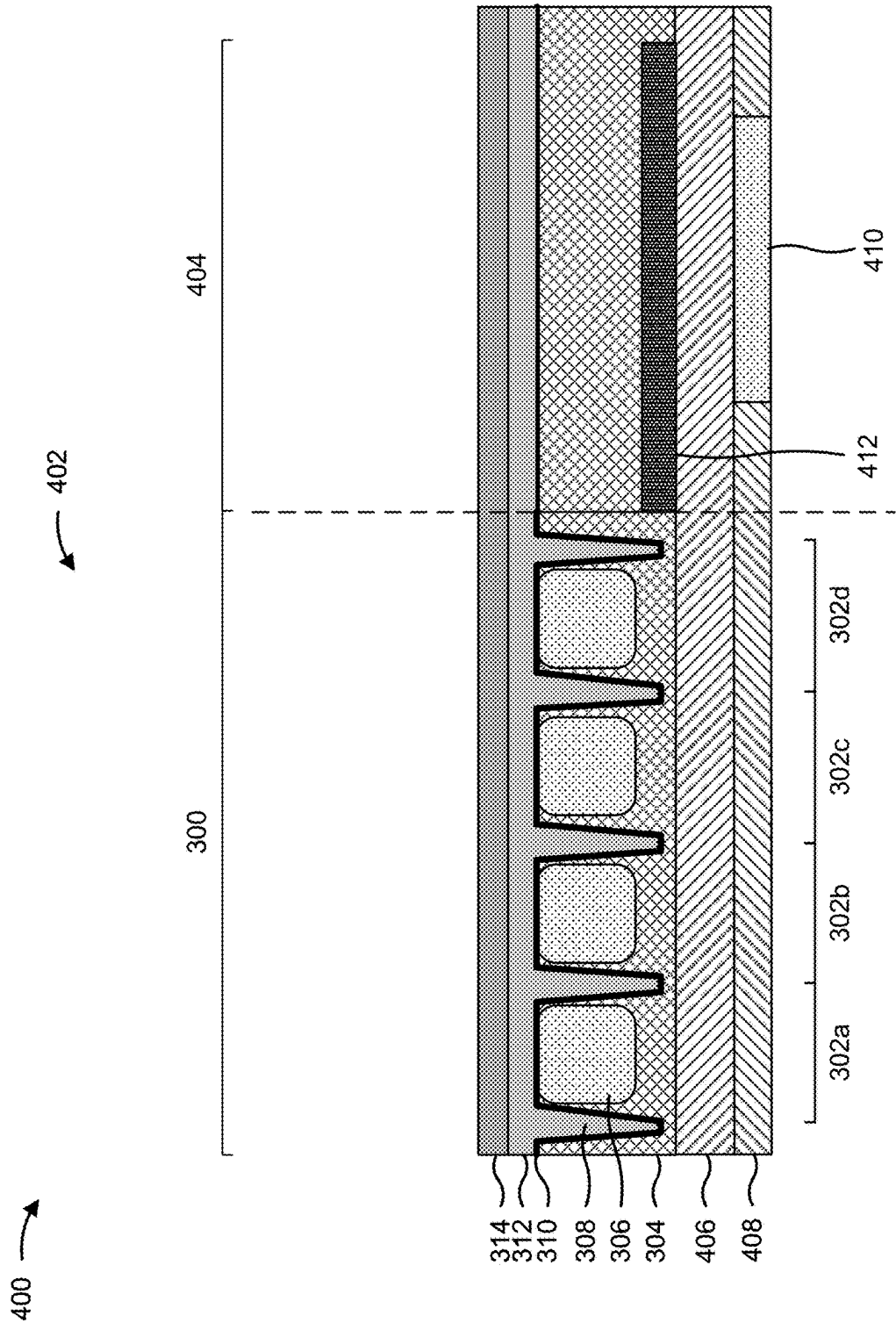


FIG. 4F

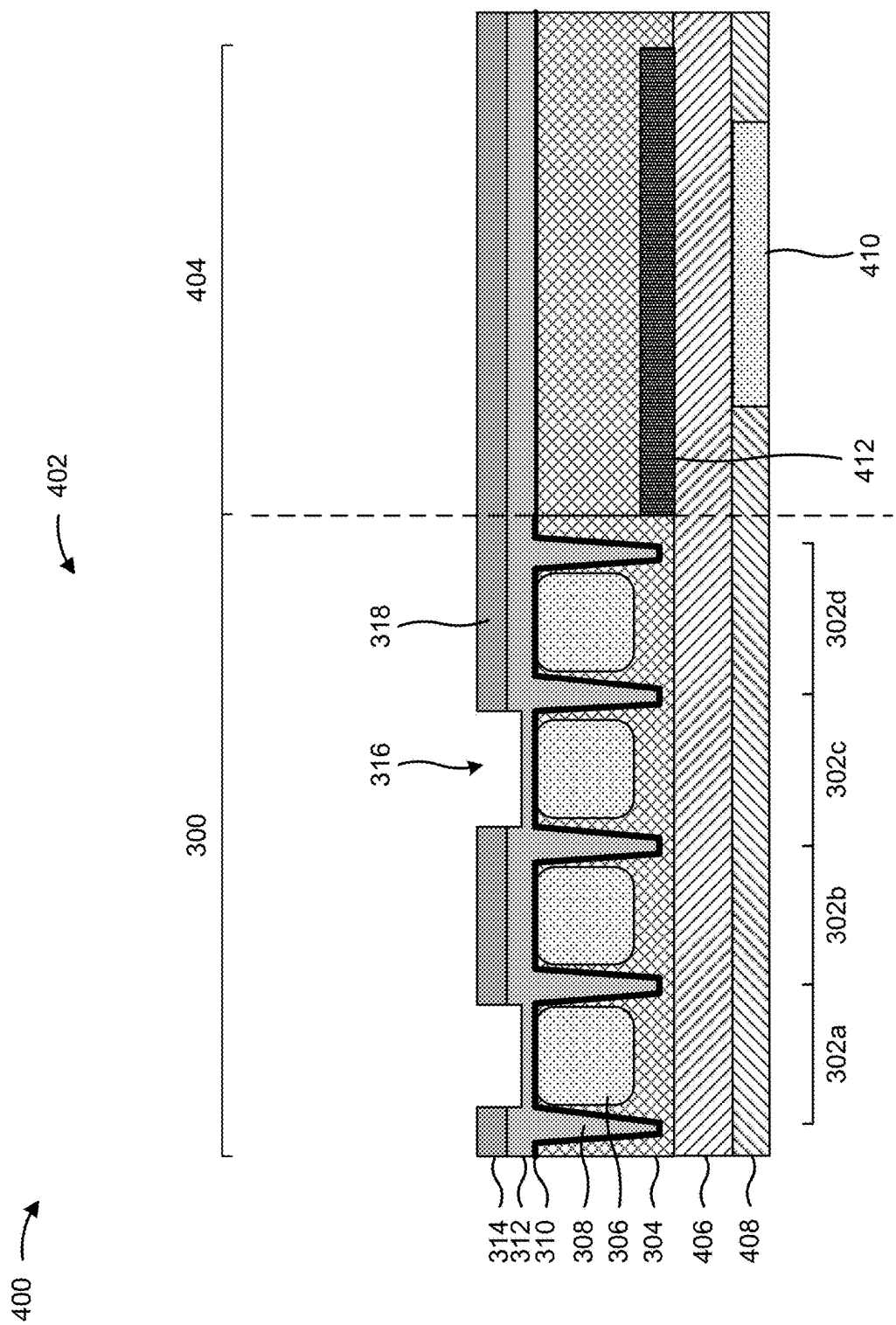


FIG. 4G

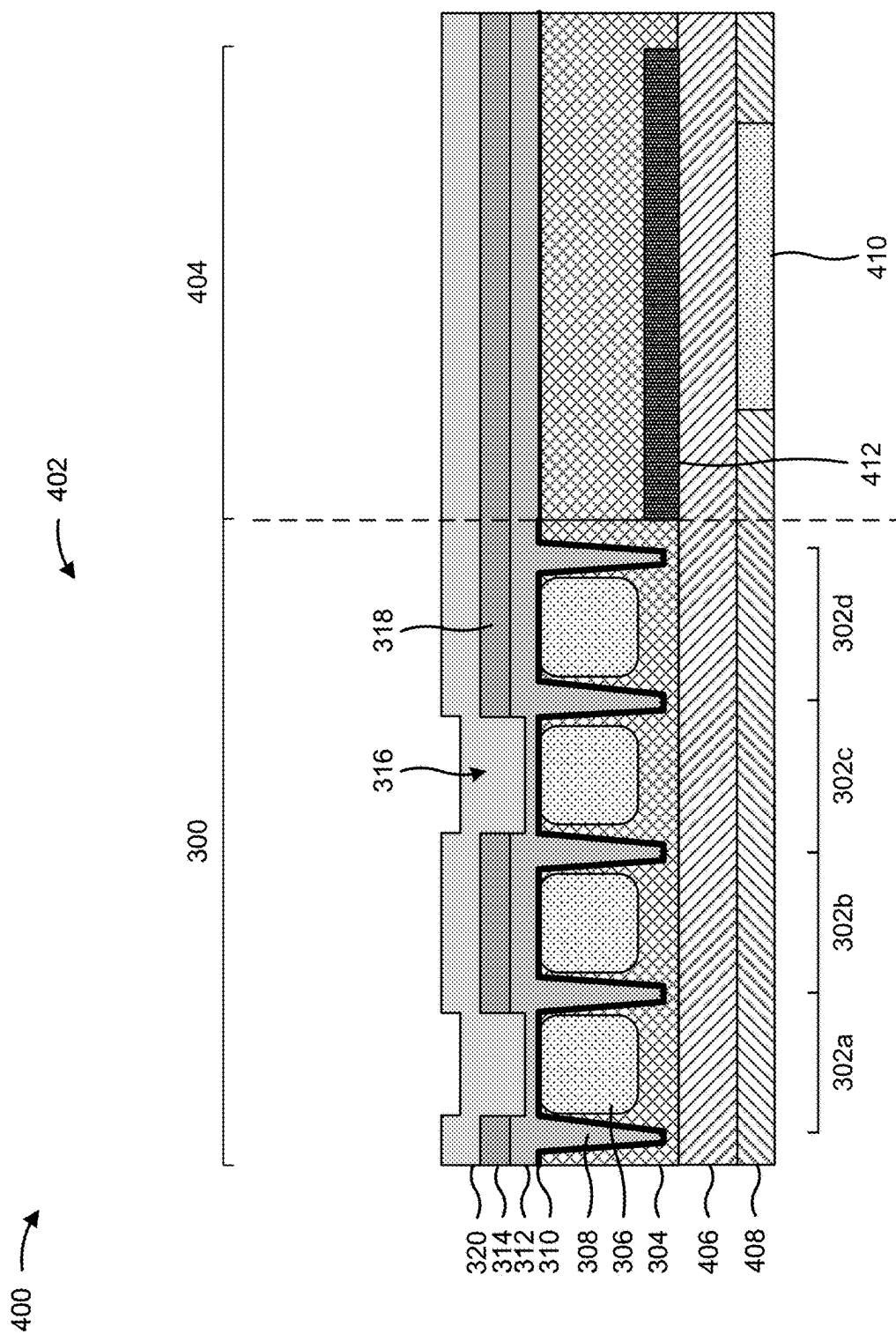


FIG. 4H

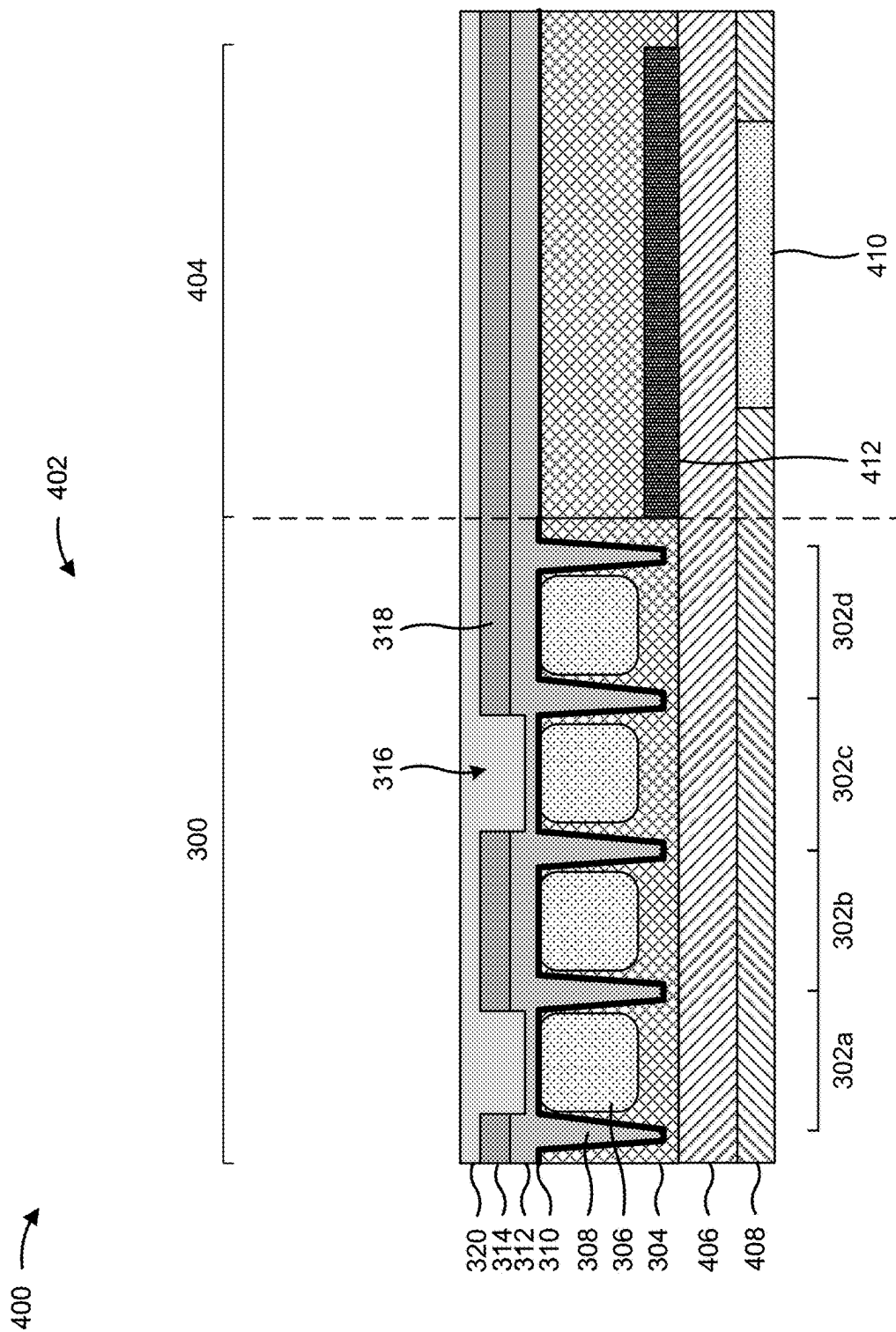
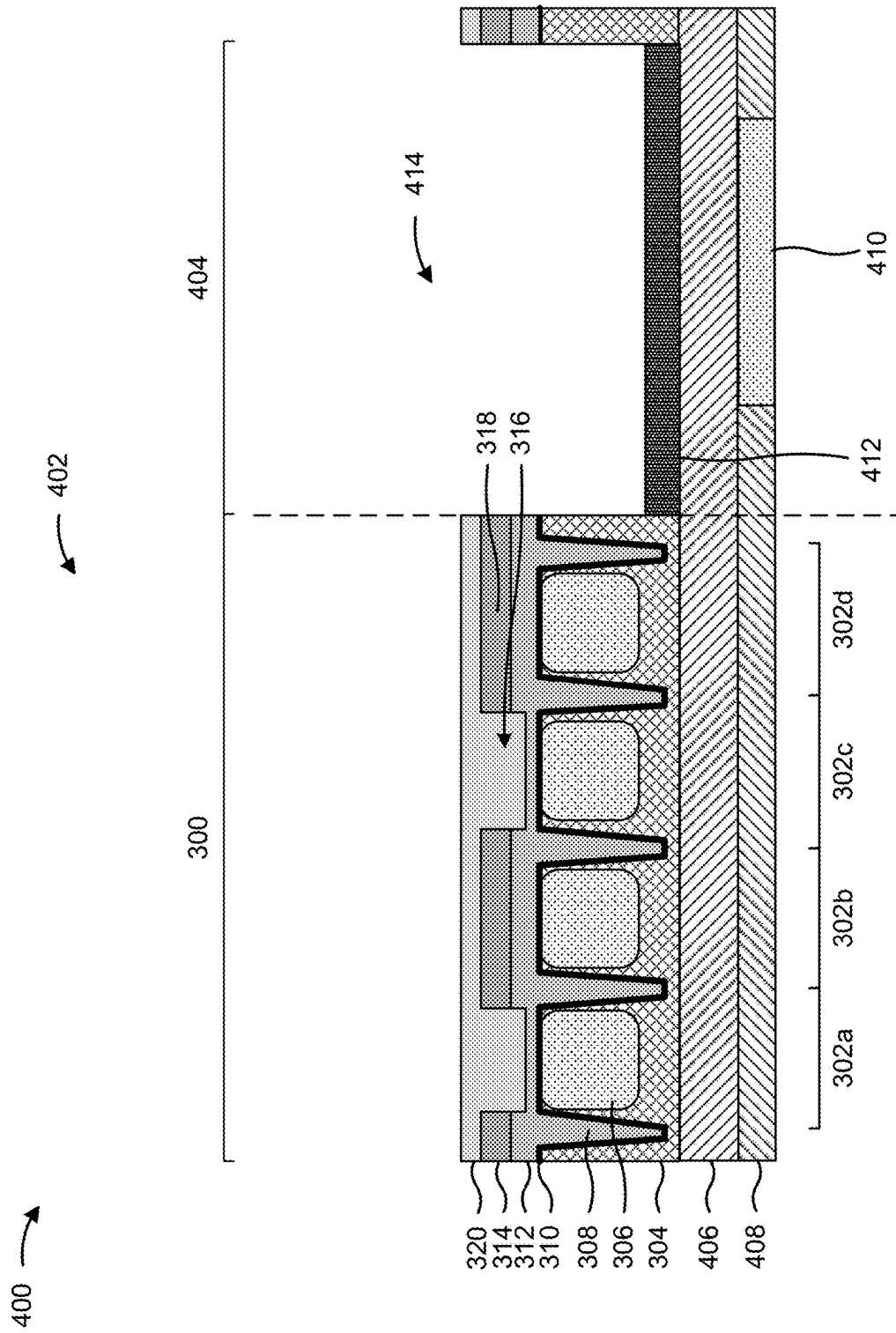


FIG. 4I



**FIG. 4J**



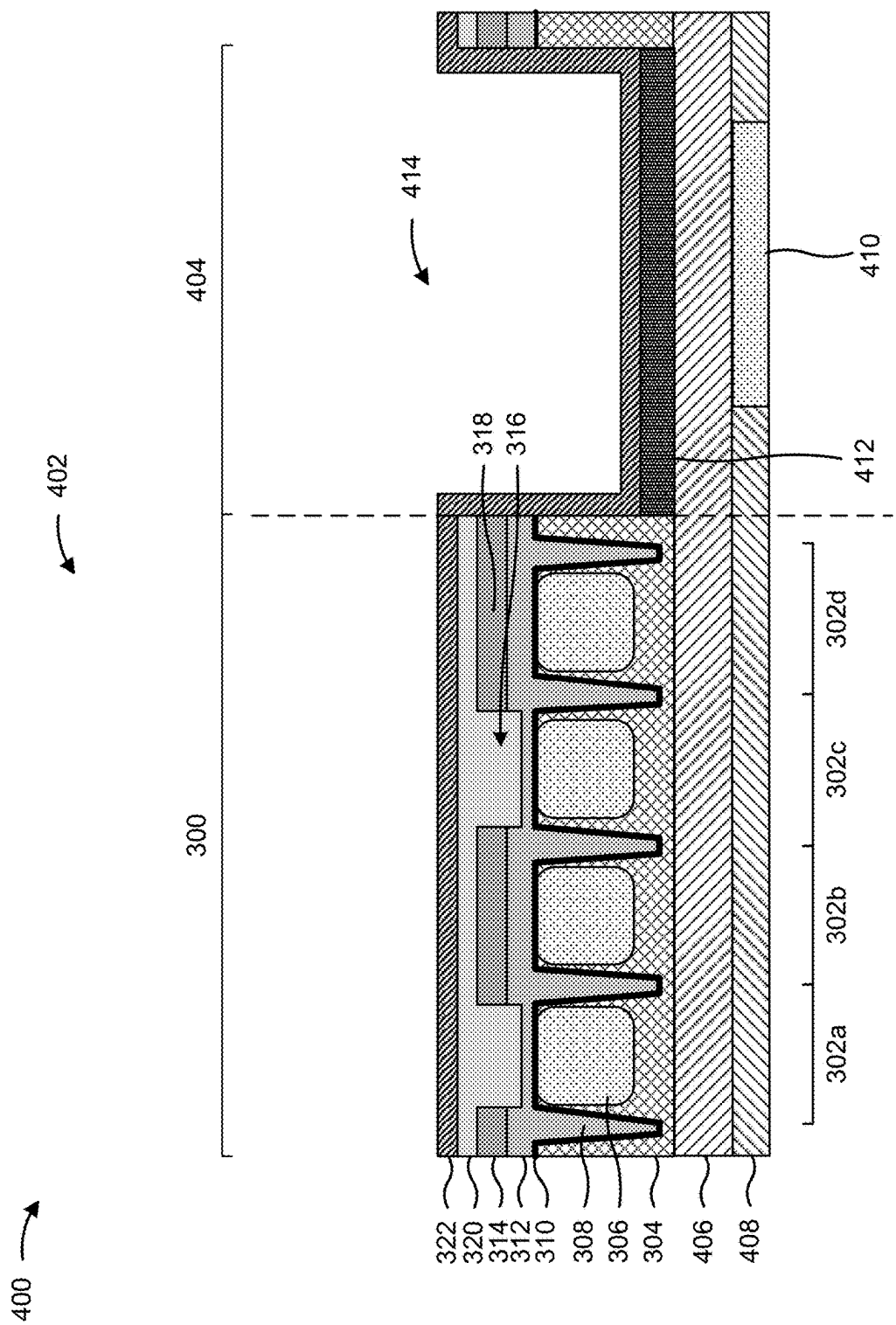


FIG. 4K

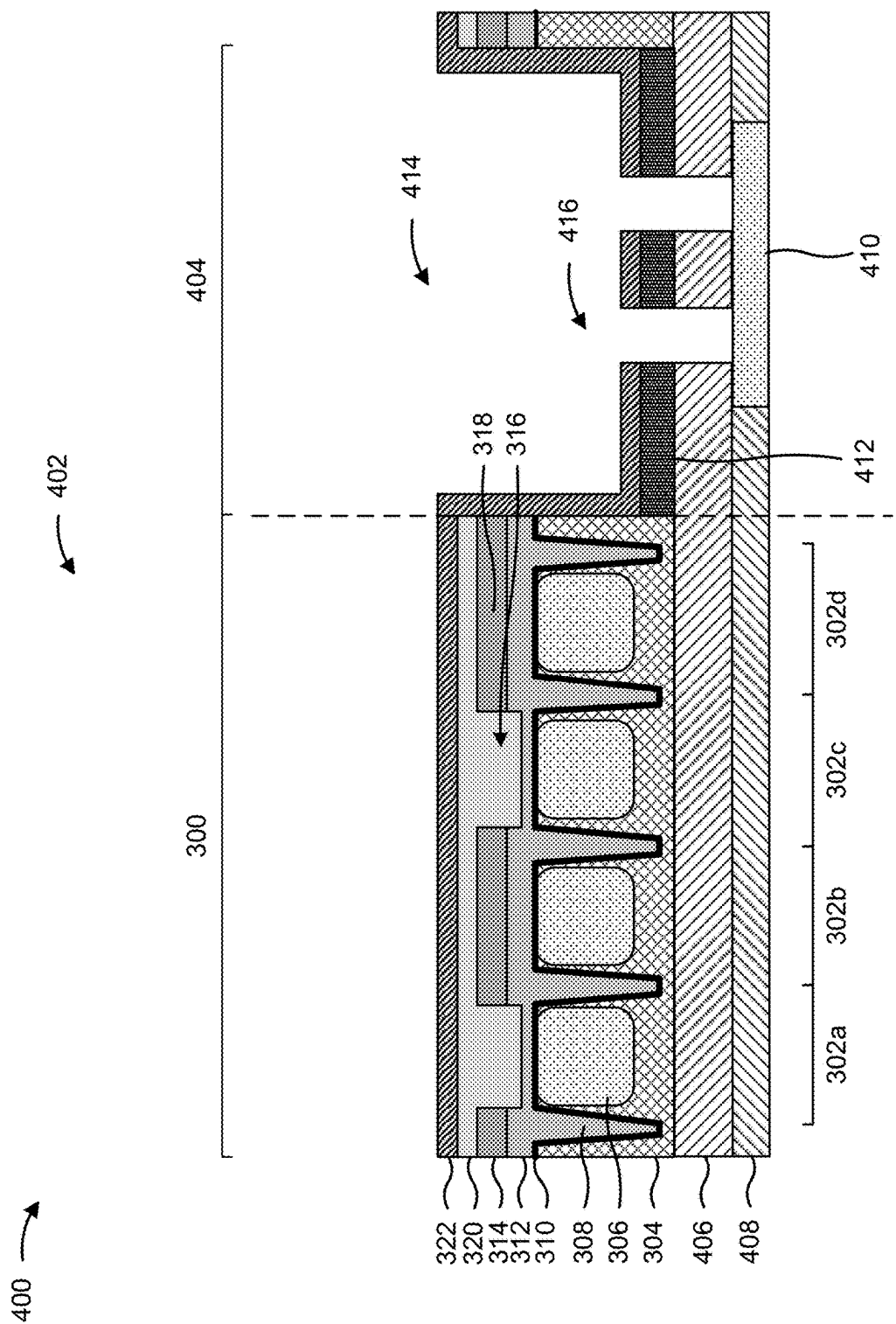


FIG. 4L

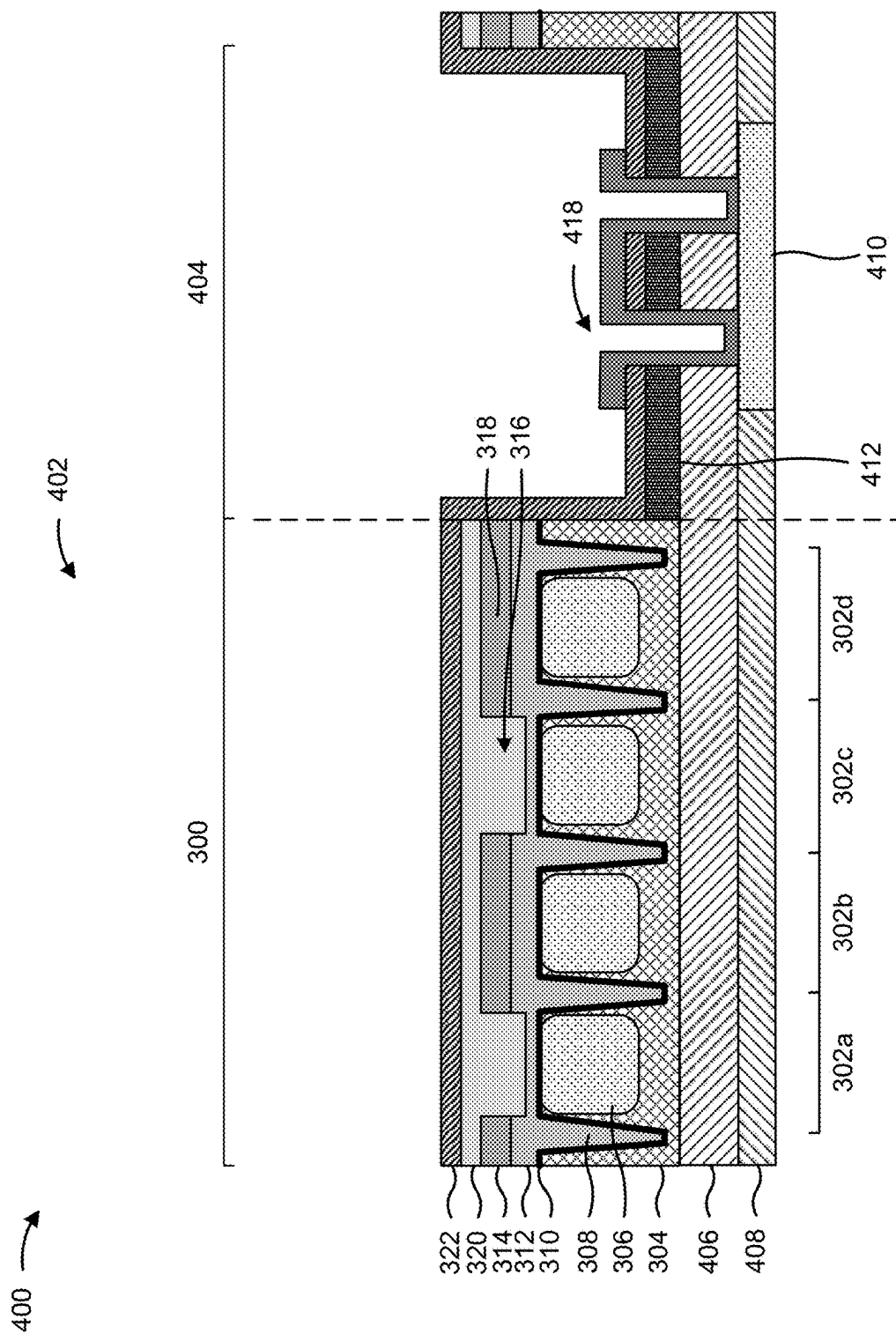


FIG. 4M

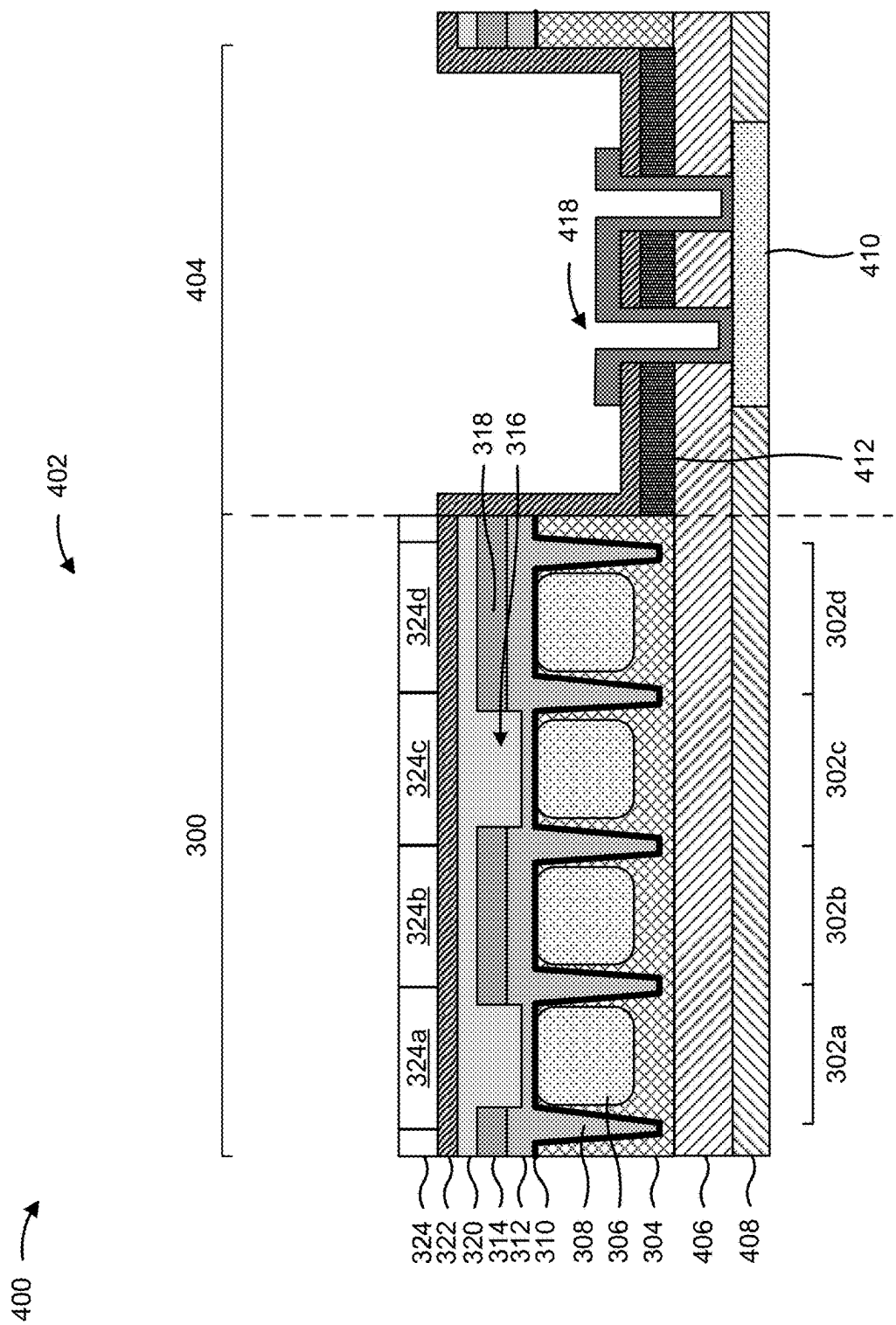


FIG. 4N

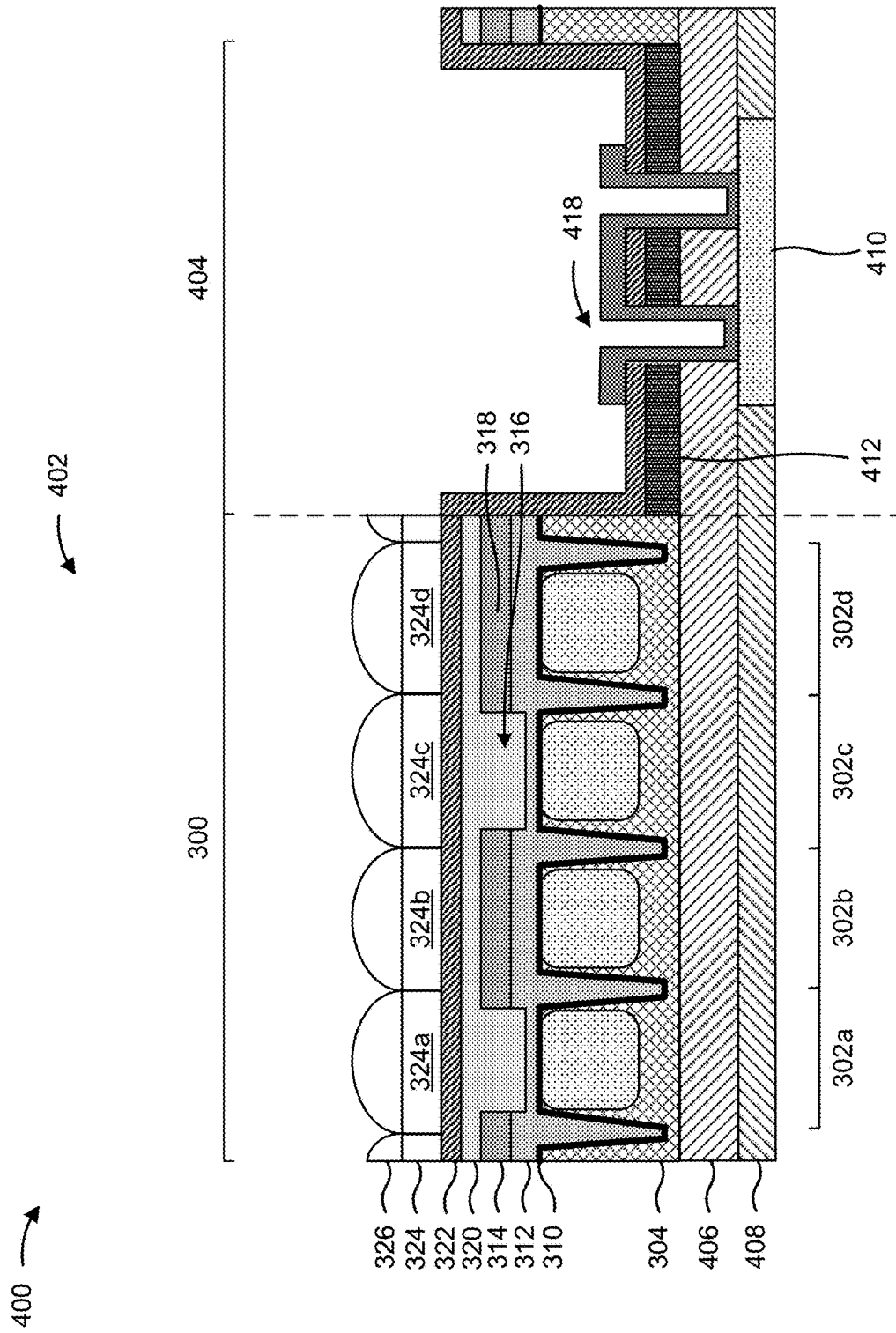


FIG. 40

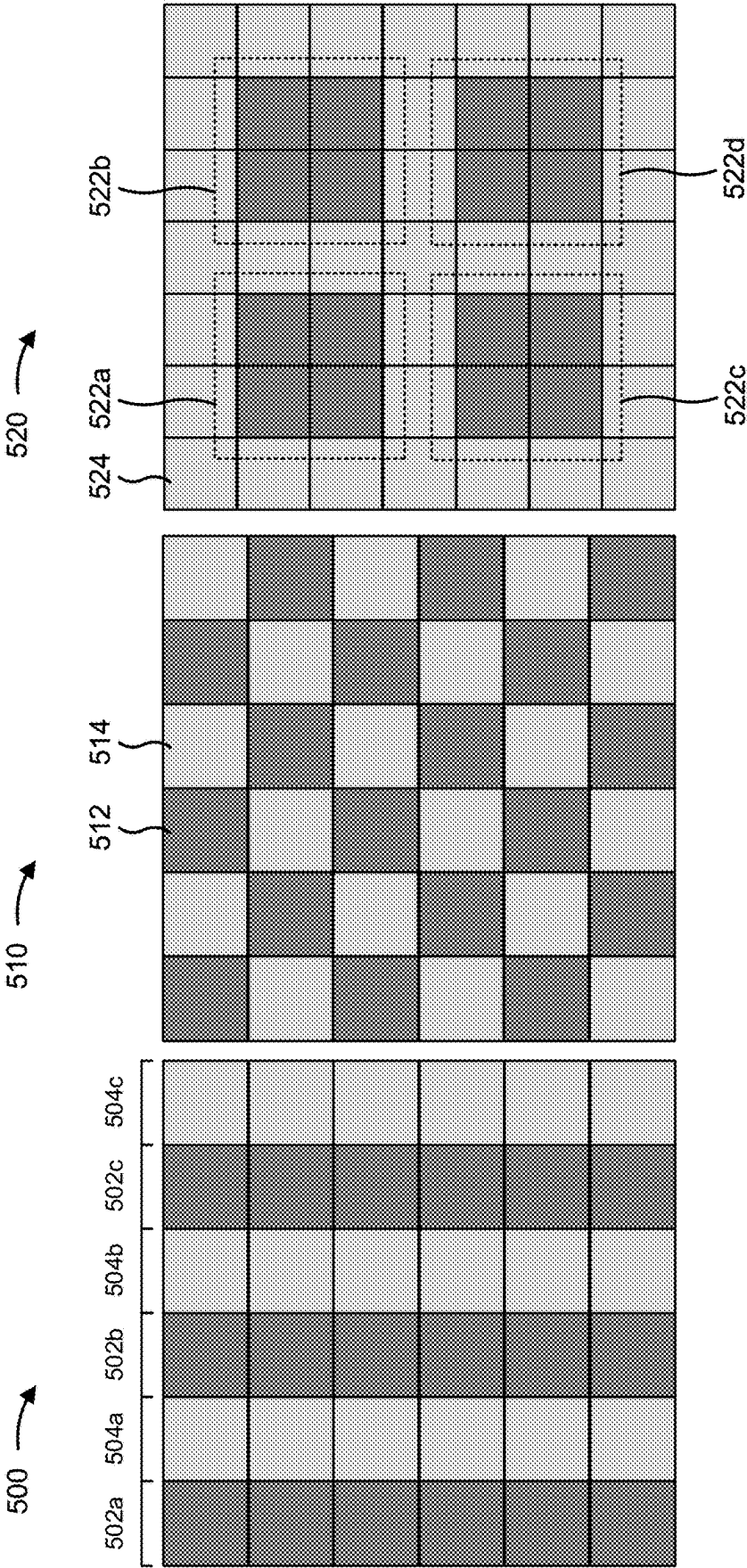


FIG. 5A

FIG. 5B

FIG. 5C

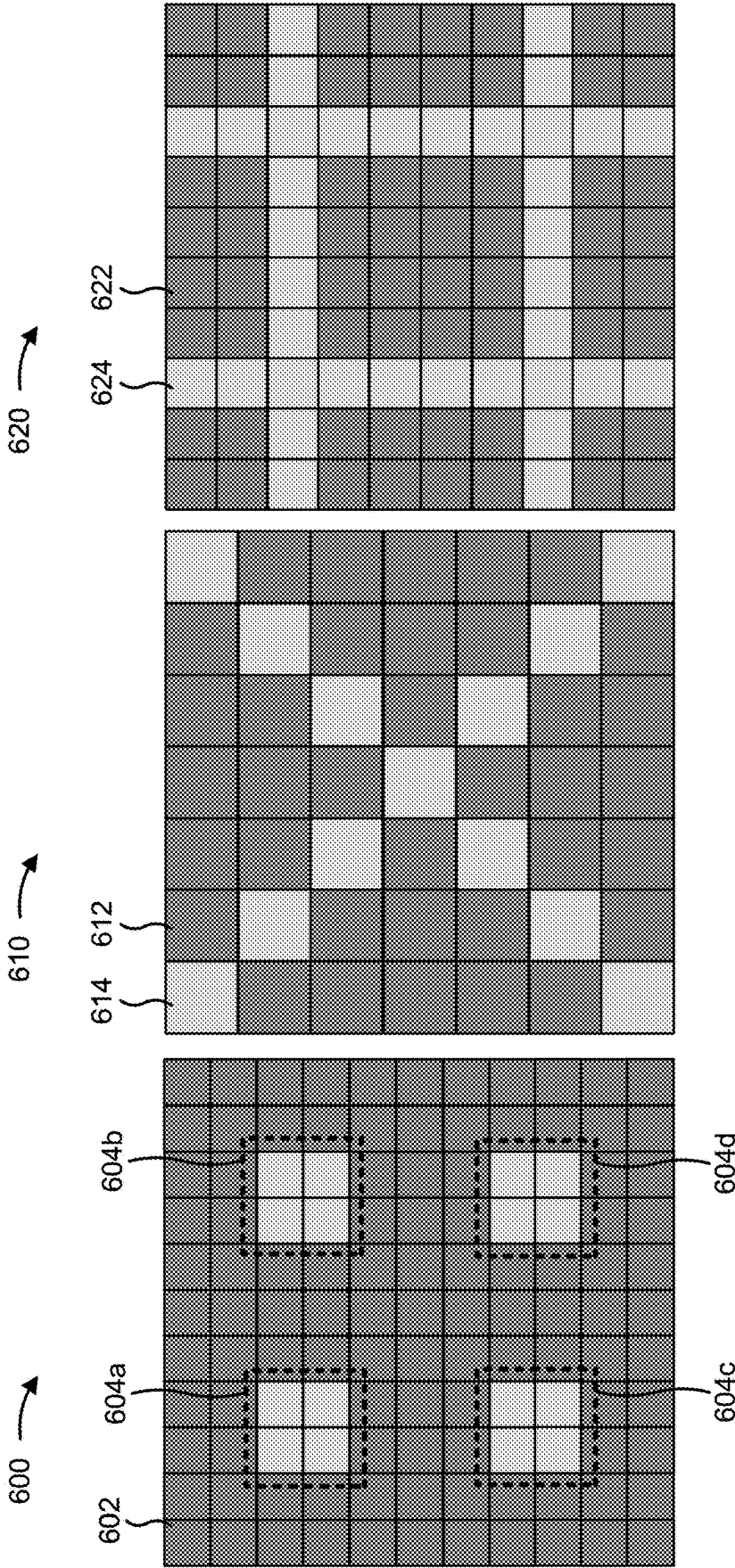


FIG. 6C

FIG. 6B

FIG. 6A

700 →

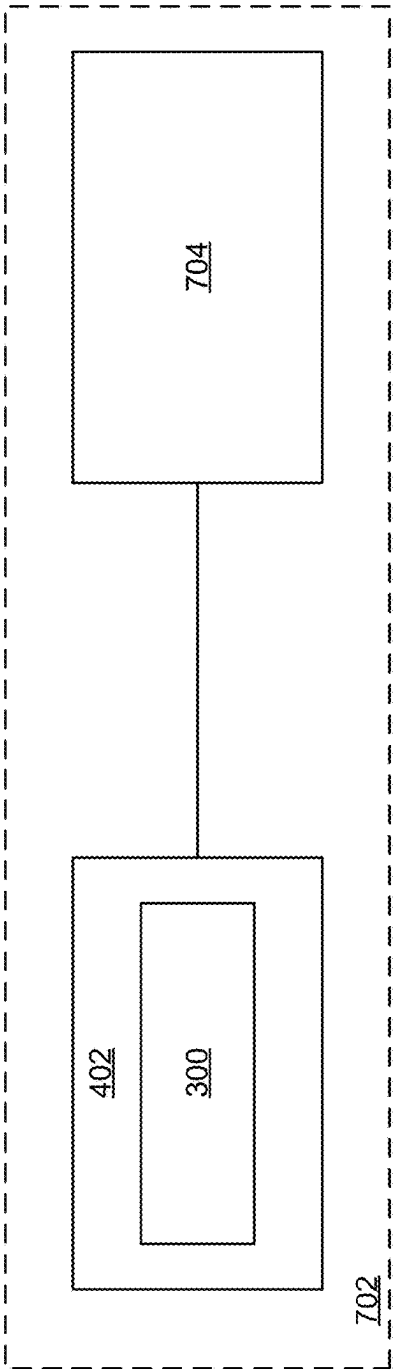


FIG. 7A



700

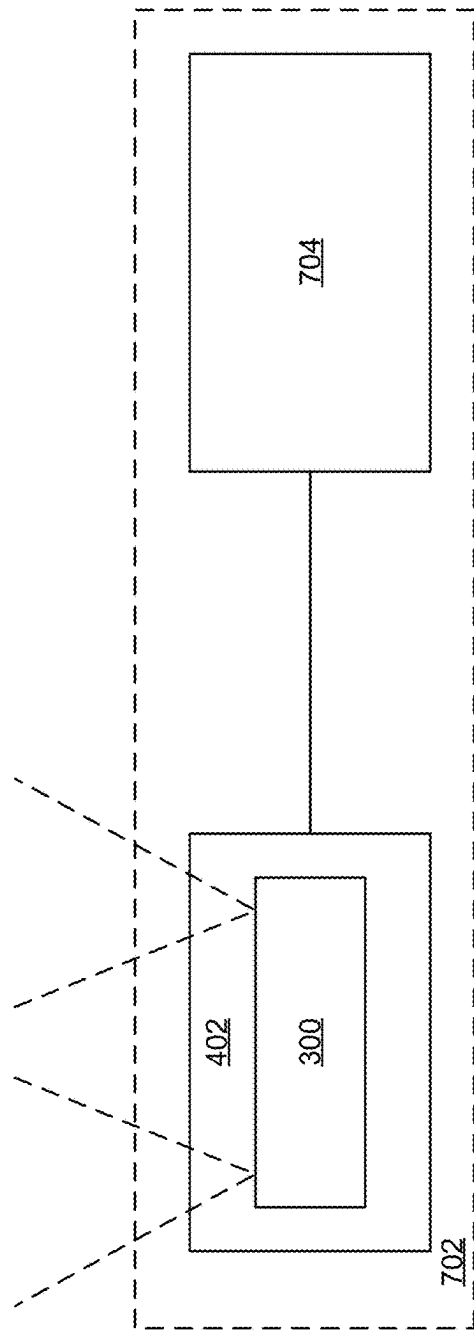
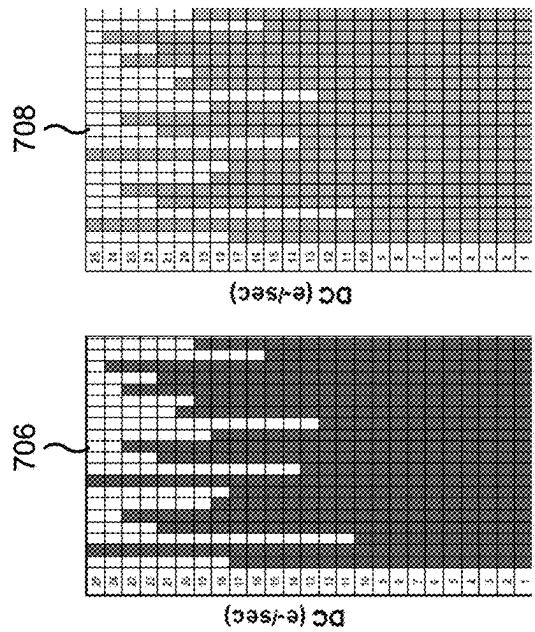


FIG. 7B

700 →

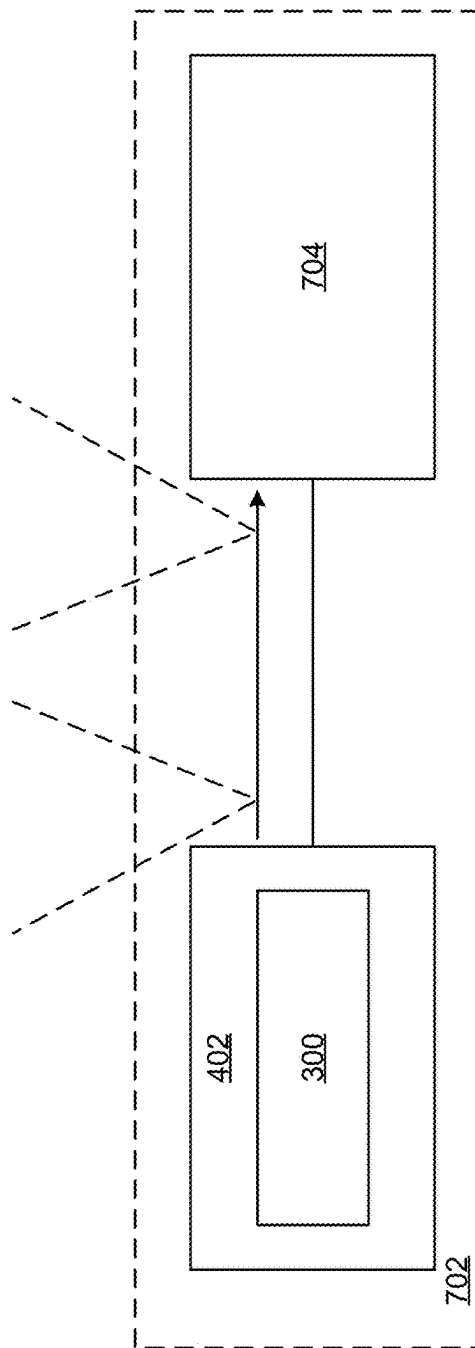
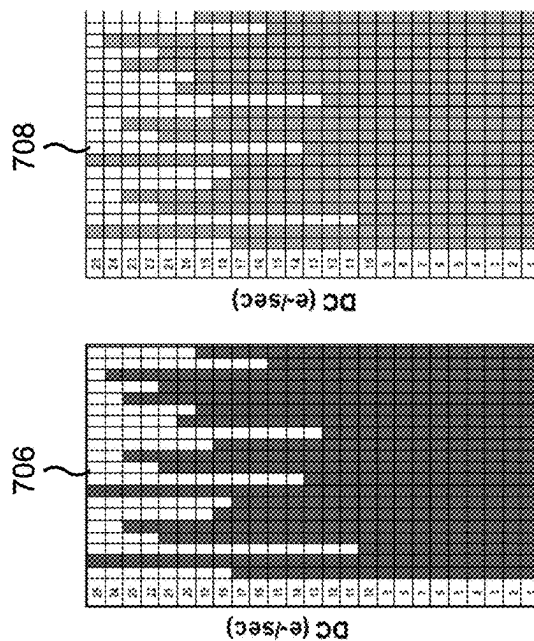


FIG. 7C

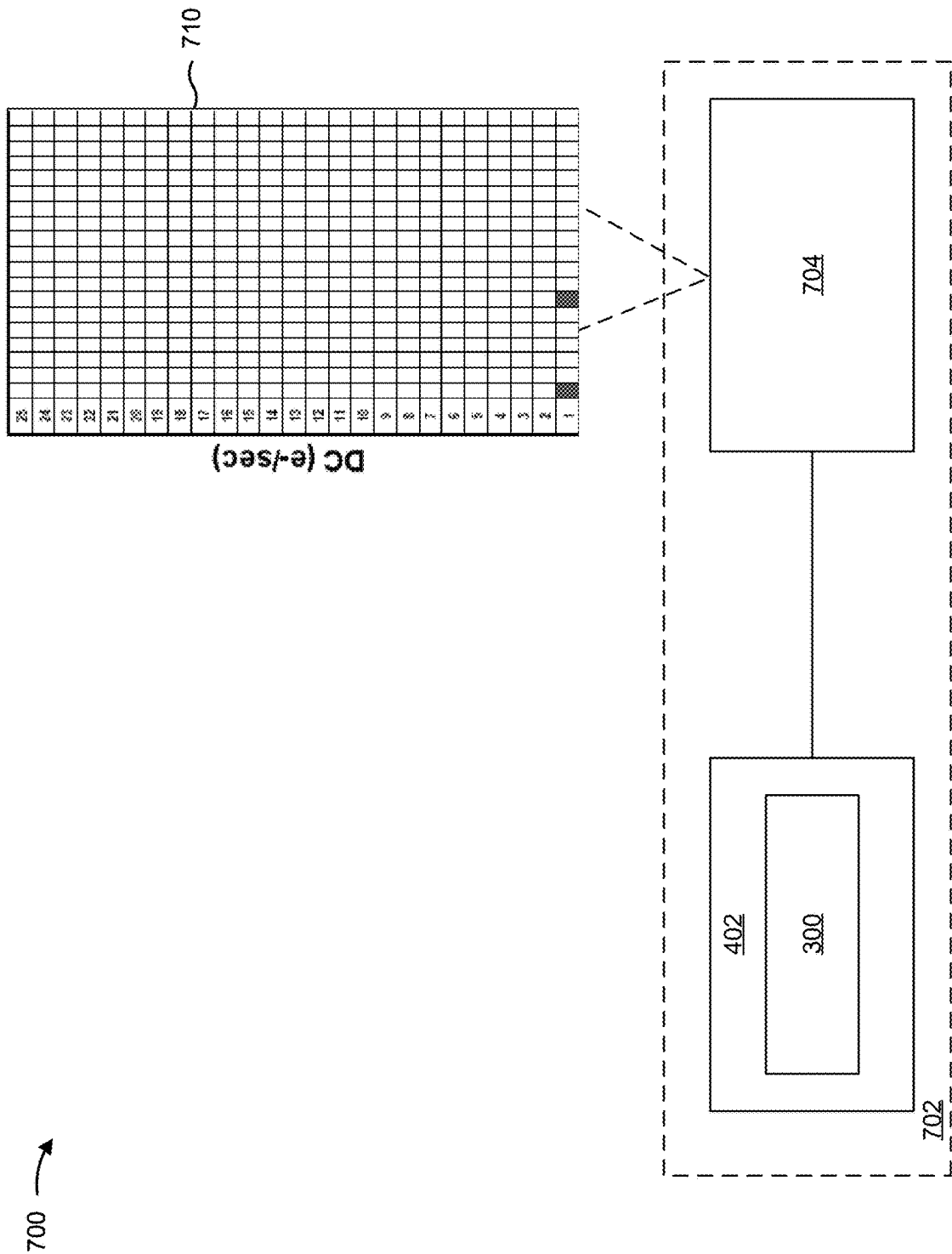


FIG. 7D

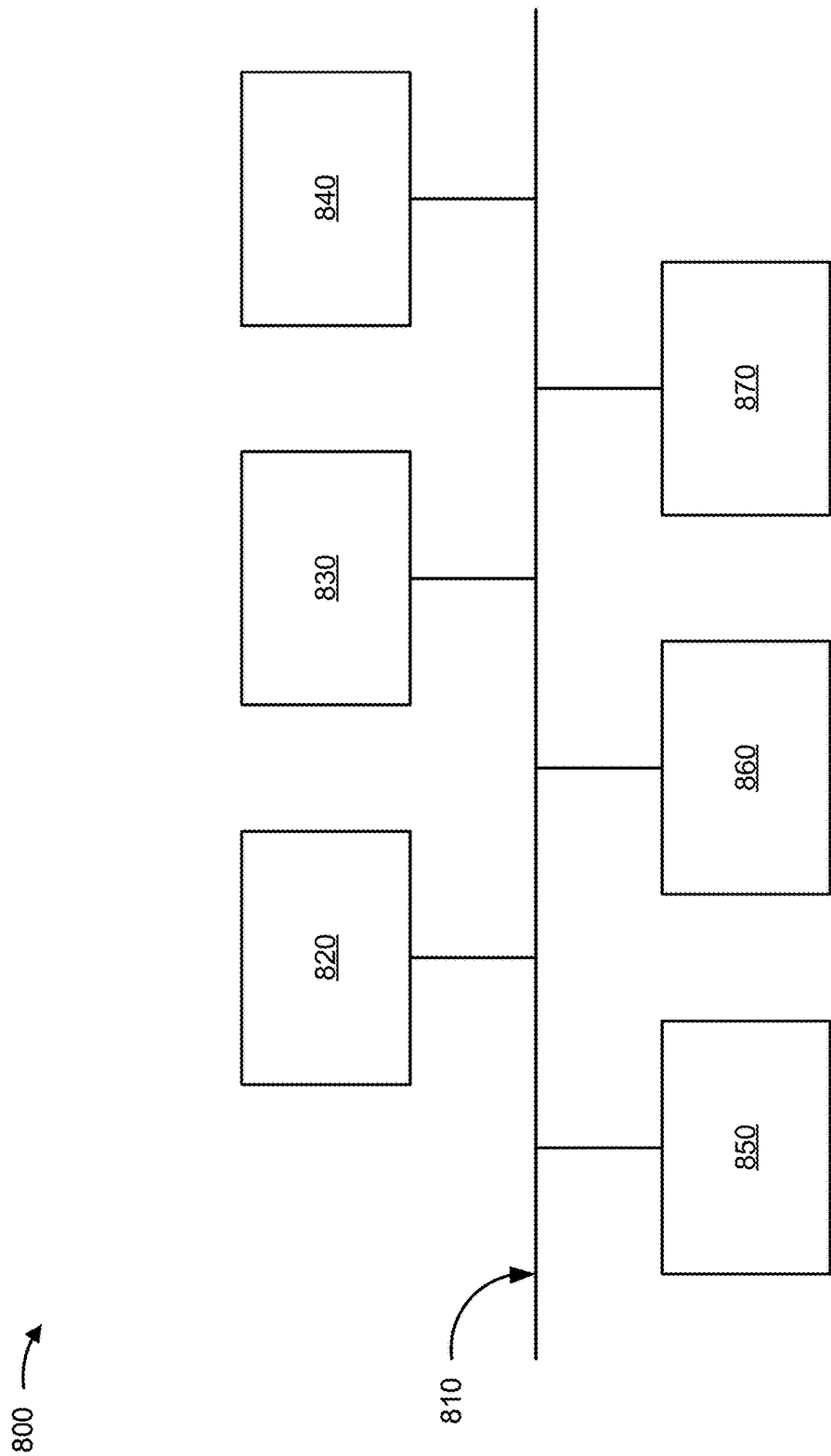


FIG. 8

900 ↗

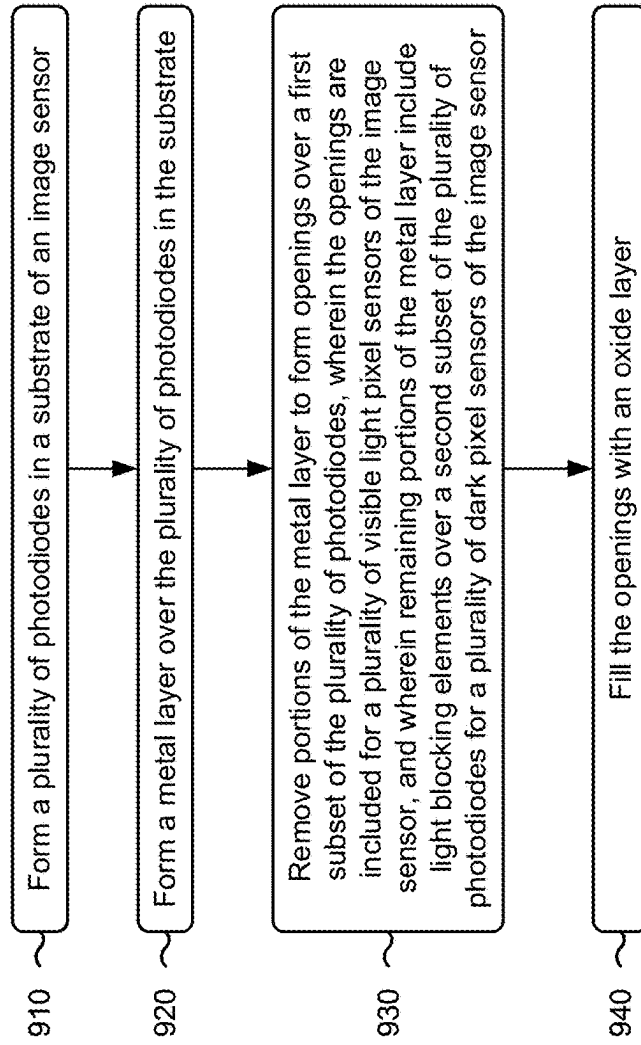


FIG. 9

1000 →

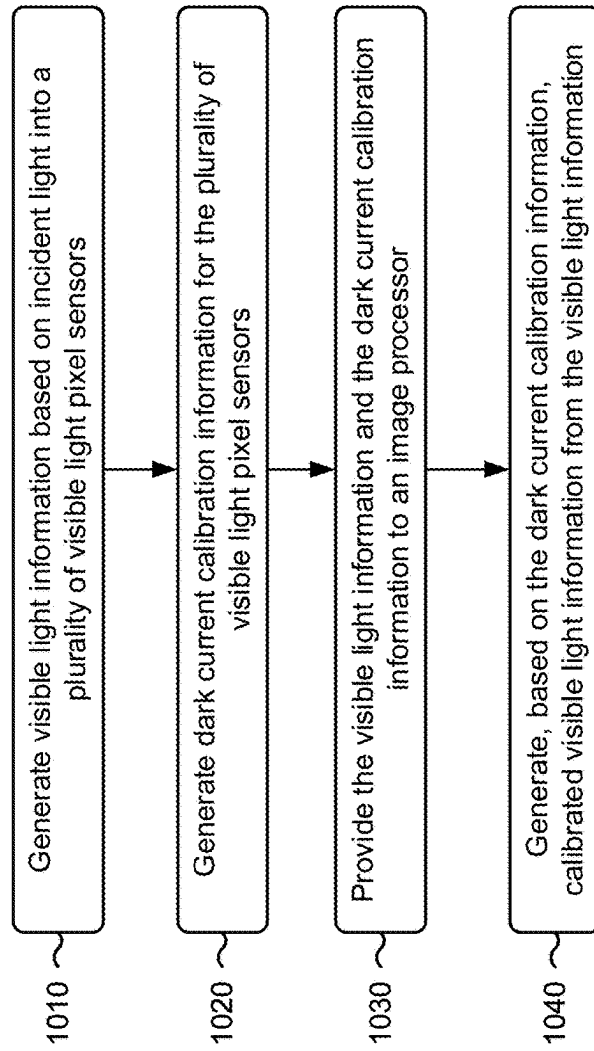


FIG. 10

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## PIXEL ARRAY INCLUDING DARK PIXEL SENSORS

### RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/249,915, filed Mar. 18, 2021, which is incorporated herein by reference in its entirety.

### BACKGROUND

Complementary metal oxide semiconductor (CMOS) image sensors utilize light-sensitive CMOS circuitry to convert light energy into electrical energy. The light-sensitive CMOS circuitry may include a photodiode formed in a silicon substrate. As the photodiode is exposed to light, an electrical charge is induced in the photodiode (referred to as a photocurrent). The photodiode may be coupled to a switching transistor, which is used to sample the charge of the photodiode. Colors may be determined by placing filters over the light-sensitive CMOS circuitry.

Light received by pixel sensors of a CMOS image sensor is often based on the three primary colors: red, green, and blue (R, G, B). Pixel sensors that sense light for each color can be defined through the use of a color filter that allows the light wavelength for a particular color to pass into a photodiode. Some pixel sensors may include a near infrared (NIR) pass filter, which blocks visible light and passes NIR light through to the photodiode.

### BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a diagram of an example environment in which systems and/or methods described herein may be implemented.

FIGS. 2 and 3A-3E are diagrams of an example pixel array described herein.

FIGS. 4A-4O are diagrams of an example implementation described herein.

FIGS. 5A-5C and 6A-6C are diagrams of example pixel sensor configurations described herein.

FIGS. 7A-7D are diagrams of an example implementation described herein.

FIG. 8 is a diagram of example components of one or more devices of FIG. 1.

FIG. 9 is a flowchart of an example process relating to forming a pixel array.

FIG. 10 is a flowchart of an example process relating to generating calibrated visible light information.

### DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in

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which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Dark current is an electrical current that occurs in a pixel sensor as a result of an energy source other than incident light. Dark current may result from, for example, heat generated by an image sensor in which the pixel sensor is included and/or one or more other devices near the image sensor. Dark current can cause noise and other defects in images and/or video captured by the image sensor. For example, dark current can artificially increase the photocurrent generated by the pixel sensors included in the image sensor, which can cause some of the pixels in an image or a video to register as a white pixel or a hot pixel.

Some implementations described herein provide a pixel array that includes a plurality of dark pixel sensors configured to generate dark current calibration information for a plurality of visible light pixel sensors included in the pixel array. In particular, the plurality of dark pixel sensors may generate respective dark current measurements for each of the plurality of visible light pixel sensors (e.g., one-to-one or per-pixel-sensor dark current measurements) or for small subsets of the plurality of visible light pixel sensors (e.g., subsets of two to four visible light pixel sensors). In this way, each of the plurality of visible light pixel sensors may be individually calibrated (or small subsets of the plurality of visible light pixel sensors may be individually calibrated) based on an estimated dark current experienced by each of the plurality of visible light pixel sensors. This individual dark current calibration may enable more accurate dark current calibration of the visible light pixel sensors included in the pixel array (e.g., relative to using an average dark current measurement generated by a large border of black level correction (BLC) surrounding the visible light pixel sensors for calibration across all of the visible light pixel sensors), and may be used to account for large differences in estimated dark currents for the visible light pixel sensors.

FIG. 1 is a diagram of an example environment 100 in which systems and/or methods described herein may be implemented. As shown in FIG. 1, environment 100 may include a plurality of semiconductor processing tools 102-114 and a wafer/die transport tool 116. The plurality of semiconductor processing tools 102-114 may include a deposition tool 102, an exposure tool 104, a developer tool 106, an etch tool 108, a planarization tool 110, a plating tool 112, an ion implantation tool 114, and/or another type of semiconductor processing tool. The tools included in example environment 100 may be included in a semicon-

ductor clean room, a semiconductor foundry, a semiconductor processing facility, and/or manufacturing facility, among other examples.

The deposition tool **102** is a semiconductor processing tool that includes a semiconductor processing chamber and one or more devices capable of depositing various types of materials onto a substrate. In some implementations, the deposition tool **102** includes a spin coating tool that is capable of depositing a photoresist layer on a substrate such as a wafer. In some implementations, the deposition tool **102** includes a chemical vapor deposition (CVD) tool such as a plasma-enhanced CVD (PECVD) tool, a high-density plasma CVD (HDP-CVD) tool, a sub-atmospheric CVD (SACVD) tool, an atomic layer deposition (ALD) tool, a plasma-enhanced atomic layer deposition (PEALD) tool, or another type of CVD tool. In some implementations, the deposition tool **102** includes a physical vapor deposition (PVD) tool, such as a sputtering tool or another type of PVD tool. In some implementations, the example environment **100** includes a plurality of types of deposition tools **102**.

The exposure tool **104** is a semiconductor processing tool that is capable of exposing a photoresist layer to a radiation source, such as an ultraviolet light (UV) source (e.g., a deep UV light source, an extreme UV light (EUV) source, and/or the like), an x-ray source, an electron beam (e-beam) source, and/or the like. The exposure tool **104** may expose a photoresist layer to the radiation source to transfer a pattern from a photomask to the photoresist layer. The pattern may include one or more semiconductor device layer patterns for forming one or more semiconductor devices, may include a pattern for forming one or more structures of a semiconductor device, may include a pattern for etching various portions of a semiconductor device, and/or the like. In some implementations, the exposure tool **104** includes a scanner, a stepper, or a similar type of exposure tool.

The developer tool **106** is a semiconductor processing tool that is capable of developing a photoresist layer that has been exposed to a radiation source to develop a pattern transferred to the photoresist layer from the exposure tool **104**. In some implementations, the developer tool **106** develops a pattern by removing unexposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by removing exposed portions of a photoresist layer. In some implementations, the developer tool **106** develops a pattern by dissolving exposed or unexposed portions of a photoresist layer through the use of a chemical developer.

The etch tool **108** is a semiconductor processing tool that is capable of etching various types of materials of a substrate, wafer, or semiconductor device. For example, the etch tool **108** may include a wet etch tool, a dry etch tool, and/or the like. In some implementations, the etch tool **108** includes a chamber that is filled with an etchant, and the substrate is placed in the chamber for a particular time period to remove particular amounts of one or more portions of the substrate. In some implementations, the etch tool **108** may etch one or more portions of the substrate using a plasma etch or a plasma-assisted etch, which may involve using an ionized gas to isotropically or directionally etch the one or more portions.

The planarization tool **110** is a semiconductor processing tool that is capable of polishing or planarizing various layers of a wafer or semiconductor device. For example, a planarization tool **110** may include a chemical mechanical planarization (CMP) tool and/or another type of planarization tool that polishes or planarizes a layer or surface of deposited or plated material. The planarization tool **110** may polish or

planarize a surface of a semiconductor device with a combination of chemical and mechanical forces (e.g., chemical etching and free abrasive polishing). The planarization tool **110** may utilize an abrasive and corrosive chemical slurry in conjunction with a polishing pad and retaining ring (e.g., typically of a greater diameter than the semiconductor device). The polishing pad and the semiconductor device may be pressed together by a dynamic polishing head and held in place by the retaining ring. The dynamic polishing head may rotate with different axes of rotation to remove material and even out any irregular topography of the semiconductor device, making the semiconductor device flat or planar.

The plating tool **112** is a semiconductor processing tool that is capable of plating a substrate (e.g., a wafer, a semiconductor device, and/or the like) or a portion thereof with one or more metals. For example, the plating tool **112** may include a copper electroplating device, an aluminum electroplating device, a nickel electroplating device, a tin electroplating device, a compound material or alloy (e.g., tin-silver, tin-lead, and/or the like) electroplating device, and/or an electroplating device for one or more other types of conductive materials, metals, and/or similar types of materials.

The ion implantation tool **114** is a semiconductor processing tool that is capable of implanting ions into a substrate. The ion implantation tool **114** may generate ions in an arc chamber from a source material such as a gas or a solid. The source material may be provided into the arc chamber, and an arc voltage is discharged between a cathode and an electrode to produce a plasma containing ions of the source material. One or more extraction electrodes may be used to extract the ions from the plasma in the arc chamber and accelerate the ions to form an ion beam. The ion beam may be directed toward the substrate such that the ions are implanted below the surface of the substrate.

Wafer/die transport tool **116** includes a mobile robot, a robot arm, a tram or rail car, an overhead hoist transport (OHT) system, an automated materially handling system (AMHS), and/or another type of device that is used to transport wafers and/or dies between semiconductor processing tools **102-114** and/or to and from other locations such as a wafer rack, a storage room, and/or the like. In some implementations, wafer/die transport tool **116** may be a programmed device that is configured to travel a particular path and/or may operate semi-autonomously or autonomously.

The number and arrangement of devices shown in FIG. 1 are provided as one or more examples. In practice, there may be additional devices, fewer devices, different devices, or differently arranged devices than those shown in FIG. 1. Furthermore, two or more devices shown in FIG. 1 may be implemented within a single device, or a single device shown in FIG. 1 may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of environment **100** may perform one or more functions described as being performed by another set of devices of environment **100**.

FIG. 2 shows a top-down view of the pixel array **200**. In some implementations, the pixel array **200** may be included in an image sensor. The image sensor may include a complementary metal oxide semiconductor (CMOS) image sensor, a back side illuminated (BSI) CMOS image sensor, or another type of image sensor. As shown in FIG. 2, the pixel array **200** may include a plurality of pixel sensors **202**. As further shown in FIG. 2, the pixel sensors **202** may be arranged in a grid. In some implementations, the pixel



sensors **202** are square-shaped (as shown in the example in FIG. 2). In some implementations, the pixel sensors **202** include other shapes such as circle shapes, octagon shapes, diamond shapes, and/or other shapes.

The pixel sensors **202** may be configured to sense and/or accumulate incident light (e.g., light directed toward the pixel array **200**). For example, a pixel sensor **202** may absorb and accumulate photons of the incident light in a photodiode. The accumulation of photons in the photodiode may generate a charge representing the intensity or brightness of the incident light (e.g., a greater amount of charge may correspond to a greater intensity or brightness, and a lower amount of charge may correspond to a lower intensity or brightness).

The pixel array **200** may be electrically connected to a back-end-of-line (BEOL) metallization stack (not shown) of the image sensor. The BEOL metallization stack may electrically connect the pixel array **200** to control circuitry that may be used to measure the accumulation of incident light in the pixel sensors **202** and convert the measurements to an electrical signal.

As indicated above, FIG. 2 is provided as an example. Other examples may differ from what is described with regard to FIG. 2.

FIG. 3A is a diagram of an example pixel array **300** described herein. In some implementations, the example pixel array **300** illustrated in FIG. 3A may include, or may be included in, the pixel array **200** (or a portion thereof). In some implementations, the pixel array **300** may be included in an image sensor. The image sensor may include a CMOS image sensor, a BSI CMOS image sensor, or another type of image sensor.

As shown in FIG. 3A, the pixel array **300** may include a plurality of adjacent pixel sensors **302**, such as pixel sensors **302a-302d**. In some implementations, the pixel sensors **302a-302d** are configured as square-shaped pixel sensors **202** included in the pixel array **200**. In some implementations, the pixel sensors **302a-302d** include other shape(s) of pixel sensors such as octagon-shaped pixel sensors or a combination of square-shaped and octagon-shaped pixel sensors.

The pixel sensors **302** may be formed in a substrate **304**, which may include a semiconductor die substrate, a semiconductor wafer, or another type of substrate in which semiconductor pixels may be formed. In some implementations, the substrate **304** is formed of silicon (Si), a material including silicon, a III-V compound semiconductor material such as gallium arsenide (GaAs), a silicon on insulator (SOI), or another type of semiconductor material that is capable of generating a charge from photons of incident light.

Each pixel sensor **302** may include a photodiode **306**. A photodiode **306** may include a region of the substrate **304** that is doped with a plurality of types of ions to form a p-n junction or a PIN junction (e.g., a junction between a p-type portion, an intrinsic (or undoped) type portion, and an n-type portion). For example, the substrate **304** may be doped with an n-type dopant to form a first portion (e.g., an n-type portion) of a photodiode **306** and a p-type dopant to form a second portion (e.g., a p-type portion) of the photodiode **306**. A photodiode **306** may be configured to absorb photons of incident light. The absorption of photons causes a photodiode **306** to accumulate a charge (referred to as a photocurrent) due to the photoelectric effect. Here, photons bombard the photodiode **306**, which causes emission of electrons of the photodiode **306**. The emission of electrons causes the formation of electron-hole pairs, where the elec-

trons migrate toward the cathode of the photodiode **306** and the holes migrate toward the anode, which produces the photocurrent.

An isolation structure **308** may be included in the substrate **304** between adjacent pixel sensors **302**. The isolation structure **308** may provide optical isolation by blocking or preventing diffusion or bleeding of light from one pixel sensor **302** to another pixel sensor **302**, thereby reducing crosstalk between adjacent pixel sensors **302**. The isolation structure **308** may include trenches or deep trench isolation (DTI) structures filled with an oxide layer **312**. The isolation structure **308** may be formed in a grid layout in which the isolation structure **308** extends around the perimeters of the pixel sensors **302** in the pixel array **300** and intersects at various locations of the pixel array **300**. In some implementations, the isolation structure **308** is formed in the backside of the substrate **304** to provide optical isolation between the pixel sensors **302**, and thus may be referred to as a backside DTI (BDTI) structure.

The top surface of the substrate **304** and the surfaces of the isolation structure **308** (e.g., the sidewalls and the bottom surfaces) may be coated with an antireflective coating (ARC) **310**. The ARC **310** may decrease reflection of incident light away from the photodiodes **306** and may increase transmission of incident light into the substrate **304** and the photodiodes **306**. The ARC **310** may include a suitable material for reducing a reflection of incident light projected toward the photodiodes **306**, such as a nitrogen-containing material or other examples.

An oxide layer **312** may be located above the substrate **304** and above and/or on the ARC **310**. Moreover, the material of the oxide layer **312** may fill the isolation structure **308**. The oxide layer **312** may function as a passivation layer between the substrate **304** and upper layers of the pixel array **300**. In some implementations, the oxide layer **312** includes an oxide material such as a silicon oxide (SiO<sub>x</sub>). In some implementations, a silicon nitride (SiN<sub>x</sub>), a silicon carbide (SiC<sub>x</sub>), or a mixture thereof, such as a silicon carbon nitride (SiCN), a silicon oxynitride (SiON), or another type of dielectric material is used in place of the oxide layer **312** as a passivation layer.

A metal layer **314** may be located above and/or on the oxide layer **312** (or portions thereof). The metal layer **314** may provide crosstalk protection to reduce optical crosstalk between pixel sensors **302** of the pixel array **300**. The metal layer **314** may reduce optical crosstalk by absorbing and/or reflecting incident light that might otherwise travel through a first pixel sensor **302** and into the photodiode **306** of a second pixel sensor **302**, resulting in inaccurate photocurrent measurements in the second pixel sensor **302**. The metal layer **314** may be formed of a metal material, such as gold, silver, aluminum, a metal alloy, or a similar type of metal.

The pixel sensors **302** of the pixel array **300** may include a plurality of visible light pixel sensors (e.g., pixel sensors **302a** and **302c**) and a plurality of dark pixel sensors (e.g., pixel sensors **302b** and **302d**). The visible light pixel sensors may be configured to collect and/or absorb photons of incident light in one or more visible light wavelengths (or wavelength ranges) corresponding to one or more colors of visible light. For example, the visible light pixel sensors may include a red light pixel sensor, a green light pixel sensor, a blue light pixel sensor, and/or a yellow light pixel sensor, among other examples. A visible light pixel sensor may be configured to generate visible light information based on photons of absorbed incident light. The visible light information may include, for example, a photocurrent that is based on the intensity of incident light absorbed in a

particular visible light wavelength (or wavelength range). Color images and/or color video may be constructed from the intensities of various colors of incident light indicated in the visible light information for a plurality of visible light pixel sensors.

Openings 316 in the metal layer 314 may be included over the photodiodes 306 of the visible light pixel sensors to permit photons of incident light to travel through the metal layer 314 (e.g., through the openings 316 in the metal layer 314) and into the photodiodes 306 of the visible light pixel sensors. For example, an opening 316 may be included over the photodiode 306 of the pixel sensor 302a (which may include a visible light pixel sensor), an opening 316 may be included over the photodiode 306 of the pixel sensor 302c (which may include a visible light pixel sensor), and so on. The width of the openings 316 may be approximately as wide as, or wider than, the width of the photodiodes 306 so as to not impact the quantum efficiency of the visible light pixel sensors.

A dark pixel sensor may refer to a pixel sensor 302 that is included in a near dark environment such that the photodiode 306 of the pixel sensor 302 is not exposed to incident light (or so that the exposure to incident light is near zero). In this way, the dark pixel sensor may be configured to generate dark current calibration information that may be used to calibrate the visible light information generated by the visible light pixel sensors. The dark current calibration information may include a measurement of a dark current that is generated based on the amount of non-light energy absorbed in a photodiode 306 (energy that may otherwise introduce noise and other defects in images and/or video generated by the pixel array 300) of a dark pixel sensor. The measurement of the dark current may be used to calibrate a photocurrent generated by a visible light pixel sensor by subtracting or removing the magnitude of the measured dark current from the photocurrent.

In some implementations, the pixel array 300 includes a dark pixel sensor for each visible light pixel sensor included in the pixel array 300. In this way, each dark pixel sensor may generate respective dark current measurements for a single respective visible light pixel sensor (e.g., a one-to-one dark current measurement or a per-pixel-sensor dark current measurement), which may be used to individually calibrate each of the visible light pixel sensors in the pixel array 300. In some implementations, the pixel array 300 includes a dark pixel sensor for a small subset of visible light pixel sensors included in the pixel array. For example, the pixel array 300 may include dark pixel sensors for each subset of two (2) visible light pixel sensors, for each subset of ten (10) visible light pixel sensors, or for each subset of twenty (20) visible light pixel sensors, among other examples.

The quantity of dark pixel sensors (and thus, the quantity of visible light pixel sensors per dark pixel sensor) included in the pixel array 300 may be configured to satisfy a dark current performance parameter, a color saturation performance parameter, a color sensitivity performance parameter, and/or one or more other types of performance parameters. As an example, the dark current performance and the dark current calibration accuracy may be increased by including a greater quantity of dark pixel sensors in the pixel array 300, such as one dark pixel sensor per visible light pixel sensor or one dark pixel sensor per 2-10 visible light pixel sensors. The quantity of dark pixel sensors may be decreased to provide an increase in dark current performance and the dark current calibration accuracy relative to using an average dark current measurement generated by a BLC region while permitting a greater area of the pixel array 300 to be

occupied by visible light pixel sensors, which may increase color saturation performance and color sensitivity performance.

The dark pixel sensors may include light blocking elements 318 over the photodiodes 306 of the dark pixel sensors to achieve the near dark environment for the photodiodes 306 of the dark pixel sensors. For example, a light blocking element 318 may be included over the photodiode 306 of the pixel sensor 302b, a light blocking element 318 may be included over the photodiode 306 of the pixel sensor 302d, and so on. The width of the light blocking elements 318 may be configured to absorb and/or reflect photons of incident light to block or prevent the photons from reaching the photodiodes 306 of the dark pixel sensors across a wide range of incident angles. For example, the width of the light blocking elements 318 may be configured such that the light blocking elements 318 fully extend between the openings 316 in the metal layer 314. In some implementations, a light blocking element 318 may be formed from the metal layer 314 and may include one or more metal layers. In some implementations, a light blocking element 318 may be formed of other types of materials, such as oxides, doped oxides, organic materials, and/or oxide derivatives, among other examples.

One or more passivation layers may be included over and/or on the metal layer 314, over and/or on the oxide layer 312 in the openings 316. For example, a BSI oxide layer 320 may be included on the metal layer 314 and on the oxide layer 312 in the openings 316. As another example, a buffer oxide layer 322 may be included on the BSI oxide layer 320 over the metal layer 314 and over the openings 316. In some implementations, the BSI oxide layer 320 and/or the buffer oxide layer 322 include an oxide material such as a silicon oxide (SiO<sub>x</sub>). In some implementations, a silicon nitride (SiN<sub>x</sub>), a silicon carbide (SiC<sub>x</sub>), or a mixture thereof, such as a silicon carbon nitride (SiCN), a silicon oxynitride (SiON), or another dielectric material is used in place of the BSI oxide layer 320 and/or the buffer oxide layer 322 as a passivation layer.

A filter layer 324 including respective filter regions for each of the pixel sensors 302 may be included over and/or on the buffer oxide layer 322. For example, a filter region 324a may be included over the photodiode 306 of the pixel sensor 302a, a filter region 324b may be included over the photodiode 306 of the pixel sensor 302b, a filter region 324c may be included over the photodiode 306 of the pixel sensor 302c, a filter region 324d may be included over the photodiode 306 of the pixel sensor 302d, and so on.

In some implementations, a filter region included in a visible light pixel sensor may include a color filter configured to filter incident light to allow a particular wavelength of the incident light to pass to a photodiode 306 of the visible light pixel sensor. For example, a filter region may filter red light for a red pixel sensor, a filter region may filter green light for a green pixel sensor, a filter region may filter blue light for a blue pixel sensor, and so on. A blue filter region may permit the component of incident light near a 450 nanometer wavelength to pass and may block other wavelengths from passing. A green filter region may permit the component of incident light near a 550 nanometer wavelength to pass and may block other wavelengths from passing. A red filter region may permit the component of incident light near a 650 nanometer wavelength to pass and may block other wavelengths from passing. A yellow filter region may permit the component of incident light near a 580 nanometer wavelength to pass and may block other wavelengths from passing.

In some implementations, a filter region may be non-discriminating, which may define a white pixel sensor. A non-discriminating filter region may include a material that permits all wavelengths of light to pass into the associated photodiode **306** (e.g., for purposes of determining overall brightness to increase light sensitivity). In some implementations, a filter region may include a near infrared (NIR) bandpass filter region, which may define an NIR pixel sensor. An NIR bandpass filter region may include a material that permits the portion of incident light in an NIR wavelength range to pass to an associated photodiode **306** while blocking visible light from passing.

A filter region included over a photodiode **306** of a dark pixel sensor (e.g., the pixel sensor **302b** and **302d**) may include an opaque and/or light-blocking material to block or prevent light (e.g., ultraviolet light, visible light, infrared light, NIR light) from reaching the photodiode **306**. The filter region may, in combination with the light blocking element **318** of the dark pixel sensor, reflect and/or absorb photons of incident light to block or prevent light from reaching the photodiode **306**.

A micro-lens layer **326** may be included above and/or on the filter layer **324**. The micro-lens layer **326** may include a respective micro-lens for each of the pixel sensors **302**. For example, a micro-lens may be formed to focus incident light toward the photodiode **306** of the pixel sensor **302a**, another micro-lens may be formed to focus incident light toward the photodiode **306** of the pixel sensor **302b**, another micro-lens may be formed to focus incident light toward the photodiode **306** of the pixel sensor **302c**, another micro-lens may be formed to focus incident light toward the photodiode **306** of the pixel sensor **302d**, and so on.

FIGS. 3B-3E illustrate various example configurations of light blocking elements **318**. As shown in FIG. 3B, a light blocking element **318** may include a layer **328** of metal material, such as gold (Au), silver (Ag), copper (Cu), aluminum (Al), another type of highly reflective metal, an alloy thereof, or a combination thereof. The metal material may absorb and/or reflect incident light away from the photodiodes **306** of the dark pixel sensors so that the photodiodes **306** of the dark pixel sensors are in a dark environment, which increases the accuracy of the dark current measurements by the dark pixel sensors. In some implementations, gold may be used in implementations where the incident light into the pixel array **300** is estimated to include shorter wavelengths of light, such as in a range of approximately 200 nanometers to approximately 600 nanometers. In some implementations, aluminum may be used in implementations where the incident light into the pixel array **300** is estimated to include relatively longer wavelengths of light, such as in a range of approximately 650 nanometers to approximately 1250 nanometers.

As shown in FIG. 3C, a light blocking element **318** may include two layers of metal material, such as the layer **328** and layer **330**. The layer **330** may include tungsten or another metal that, in combination with the metal material of the layer **328**, further increases the light blocking performance of the light blocking element **318**. In some implementations, the layer **328** is included over and/or on the layer **330**. In some implementations, the layer **330** is included over and/or on the layer **328**.

As shown in FIG. 3D, a light blocking element **318** may include three layers of metal material, such as the layer **328**, the layer **330**, and the layer **332**. The layer **332** may include a metal material such as gold (Au), silver (Ag), copper (Cu), aluminum (Al), another type of highly reflective metal, an alloy thereof, or a combination thereof. In some implemen-

tations, the layer **328** and the layer **332** may include the same metal material. In some implementations, the layer **328** and the layer **332** may include different metal materials. The layer **328** and the layer **332** may include different metal materials to increase the light blocking performance of the light blocking element **318** across a wider range of light wavelengths. As an example, the layer **328** may include gold to block shorter wavelengths of light, and the layer **332** may include aluminum to block longer wavelengths of light. In some implementations, the layer **328** is included over and/or on the layer **332**, and the layer **332** is included over and/or on the layer **330**. In some implementations, the layer **330** is included in between the layer **328** and the layer **332**. In some implementations, a light blocking element **318** may include four or more layers of metal material to further increase the light blocking performance of the light blocking element **318**, which increases the height/thickness of the light blocking element **318**.

As shown in FIG. 3E, a light blocking element **318** may include a plurality of layers of oxide compounds, nitride compounds, compound derivatives, doped compounds, and/or organic materials arranged in an alternating configuration. The alternating configuration may include alternating layers of a high refractive index material and a low refractive index material. For example, the light blocking element **318** may include a first layer **334a** of a high refractive index material, a first layer **336a** of a low refractive index material over and/or on the first layer **334a**, a second layer **334b** of a high refractive index material over and/or on the first layer **336a**, and a second layer **336b** of a low refractive index material over and/or on the second layer **334b**.

Alternating layers of high refractive index material and low refractive index material may increase the likelihood that incident light will be reflected due to the difference in the refractive indexes of the high refractive index material and the low refractive index material. In some implementations, the difference in the refractive indexes of the high refractive index material and the low refractive index material may be in a range of approximately 2% difference to approximately 20% difference. Increasing the quantity of pairs of high refractive index material and low refractive index material may increase the light blocking performance of the light blocking element **318**, but may also increase the overall height of the light blocking element **318**. A greater quantity of pairs of high refractive index material and low refractive index material may be included in the light blocking element **318** in implementations where the difference in the refractive indexes of the high refractive index material and the low refractive index is relatively small (e.g., 2% difference) to account for the relatively small difference. Fewer pairs of high refractive index material and low refractive index material may be included in the light blocking element **318** in implementations where the difference in the refractive indexes of the high refractive index material and the low refractive index is relatively large (e.g., 20% difference), as fewer pairs may achieve a light blocking performance parameter for the light blocking element **318**.

Example compounds that may be included in a light blocking element **318** include, but are not limited to, silicon oxide (SiO), silicon dioxide (SiO<sub>2</sub>), magnesium oxide (MgO), aluminum oxide (Al<sub>2</sub>O<sub>3</sub>), ytterbium(III) oxide (Yb<sub>2</sub>O<sub>3</sub>), zinc oxide (ZnO), silicon nitride (Si<sub>3</sub>N<sub>4</sub>), tantalum (V) oxide (Ta<sub>2</sub>O<sub>5</sub>), zirconium dioxide (ZrO<sub>2</sub>), hafnium dioxide (HfO<sub>2</sub>), tellurium dioxide (TeO<sub>2</sub>), and/or titanium dioxide (TiO<sub>2</sub>). Example compound derivatives may include, but are not limited to, derivatives of the compounds described above that include calcium fluoride (CaF<sub>2</sub>), boron (B),

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barium (Ba), phosphorous (P), lanthanum (La), and/or lead (Pb). Example doped compounds may include, but are not limited to, the compounds described above doped with silicon (Si), oxygen (O), and/or nitrogen (Ni).

The thickness of each of the layers **328-336** may be equal to or less than approximately 5000 angstroms to reduce and/or minimize stress and peeling of the layers **328-336**. Other values of the thickness of the layers **328-336** are within the scope of the present disclosure.

As indicated above, FIGS. **3A-3E** are provided as examples. Other examples may differ from what is described with regard to FIGS. **3A-3E**.

FIGS. **4A-4O** are diagrams of an example implementation **400** described herein. Example implementation **400** may be an example process for forming an image sensor **402** including the pixel array **300**, in which the pixel array **300** includes a plurality of visible light pixel sensors and a plurality of dark pixel sensors configured to generate dark current calibration information for calibrating the visible light information generated by the plurality of visible light pixel sensors. The image sensor **402** may include a CMOS image sensor, a BSI CMOS image sensor, or another type of image sensor.

As shown in FIG. **4A**, the image sensor **402** may include a plurality of regions, such as the pixel array **300** and a bonding pad region **404** (which may also be referred to as an E-pad region), among other examples. Moreover, the image sensor **402** may include a plurality of layers, including the substrate **304**, an interlayer dielectric (ILD) layer **406** on the substrate **304**, and an inter-metal dielectric (IMD) layer **408** on the ILD layer **406**, among other examples.

The pixel array **300** may include the pixel sensors **302** of the image sensor **402**, such as the pixel sensors **302a-302d**. The bonding pad region **404** may include one or more conductive bonding pads **410** (or E-pads) in the IMD layer **408**, through which electrical connections between measurement circuitry of the image sensor **402**, the pixel sensors **302**, and/or outside devices may be established. Moreover, the bonding pad region **404** may include a shallow trench isolation (STI) structure **412** to provide electrical isolation in the bonding pad region **404**.

The ILD layer **406** may provide electrical and optical isolation between the IMD layer **408** and the substrate **304**. The IMD layer **408** may include the bonding pads **410** and other metal interconnecting structures that connect the image sensor **402** to a package, to external electrical connections, and/or to other external devices.

As shown in FIG. **4B**, one or more semiconductor processing tools may form a plurality of photodiodes **306** in the substrate **304**. For example, the ion implantation tool **114** may dope the portions of the substrate **304** using an ion implantation technique to form a respective photodiode **306** for each of the pixel sensors **302**, such as the pixel sensor **302a**, the pixel sensor **302b**, the pixel sensor **302c**, and the pixel sensor **302d**. The substrate **304** may be doped with a plurality of types of ions to form a p-n junction for each photodiode **306**. For example, the substrate **304** may be doped with an n-type dopant to form a first portion (e.g., an n-type portion) of a photodiode **306** and a p-type dopant to form a second portion (e.g., a p-type portion) of the photodiode **306**. In some implementations, another technique is used to form the photodiodes **306** such as diffusion.

As shown in FIG. **4C**, the isolation structure **308** may be formed in the substrate **304**. In particular, the isolation structure **308** may be formed between each of the photodiodes **306** of the pixel sensors **302**. In some implementations, one or more semiconductor processing tools may be used to form the isolation structure **308** in the substrate **304**.

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For example, the deposition tool **102** may form a photoresist layer on the top surface of the substrate **304**, the exposure tool **104** may expose the photoresist layer to a radiation source to pattern the photoresist layer, the developer tool **106** may develop and remove portions of the photoresist layer to expose the pattern, and the etch tool **108** may etch the portions of substrate **304** to form the isolation structure **308** in the substrate **304**. The etch tool **108** may etch the isolation structure **308** from the top surface of the substrate **304** and into the substrate **304** to form the isolation structure **308**. In some implementations, a photoresist removal tool removes the remaining portions of the photoresist layer (e.g., using a chemical stripper and/or another technique) after the etch tool **108** etches the substrate **304**.

As shown in FIG. **4D**, the ARC **310** may be formed over and/or on the top surface of the substrate **304**. Moreover, the ARC **310** may be formed in the isolation structure **308** (e.g., on the sidewalls and on the bottom surface of the isolation structure **308**). In particular, a semiconductor processing tool (e.g., the deposition tool **102**) may conformally deposit the ARC **310** using various PVD techniques, CVD techniques, and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD. In some implementations, the semiconductor processing tool may form the ARC **310** to a thickness in a range from approximately 200 angstroms to approximately 1000 angstroms.

As shown in FIG. **4E**, the isolation structure **308** may be filled with an oxide material such that the oxide layer **312** is formed in the isolation structure **308**. Moreover, the oxide layer **312** may be formed over and/or on the ARC **310**. In particular, a semiconductor processing tool (e.g., the deposition tool **102**) may deposit the oxide material such that the oxide layer **312** is formed in the isolation structure **308** and over and/or the ARC **310**. The semiconductor processing tool may deposit the oxide material using various PVD techniques, CVD techniques, and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD. In some implementations, the planarization tool **110** planarizes the oxide layer **312** after the oxide layer **312** is deposited.

As shown in FIG. **4F**, the metal layer **314** may be formed over and/or on the oxide layer **312**. In some implementations, a semiconductor processing tool (e.g., the plating tool **112**) may form the metal layer **314** using a plating technique such as electroplating (or electro-chemical deposition) or electro-less plating. In some implementations, another semiconductor processing tool (e.g., the deposition tool **102**) deposits a seed layer of the material of the metal layer **314** over and/or on the oxide layer **312** to promote adhesion between the metal layer **314** and the oxide layer **312**. The remaining material of the metal layer **314** may be deposited onto the seed layer by a plating technique.

As shown in FIG. **4G**, the openings **316** may be formed through the metal layer **314** and into a portion of the oxide layer **312** for the visible light pixel sensors (e.g., the pixel sensors **302a** and **302c**). The opening **316** may be formed by coating the metal layer **314** with a photoresist (e.g., using the deposition tool **102**), forming a pattern in the photoresist by exposing the photoresist to a radiation source (e.g., using the exposure tool **104**), removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool **106**), and etching the opening **316** into the metal layer **314** and into the portion of the oxide layer **312** (e.g., using the etch tool **108**) based on the pattern in the photoresist. The un-etched or unremoved portions of the metal layer **314** may remain over the photodiodes **306** in the dark pixel sensors (e.g., the pixel sensors **302b** and **302d**) as the light blocking elements **318**.

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As shown in FIG. 4H, the BSI oxide layer 320 may be formed in the openings 316, over and/or on the metal layer 314, and over and/or on the light blocking elements 318. In particular, a semiconductor processing tool (e.g., the deposition tool 102) may deposit the BSI oxide layer 320 using various PVD techniques, CVD techniques, and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD.

As shown in FIG. 4I, the BSI oxide layer 320 may be planarized. In particular, a semiconductor processing tool (e.g., the planarization tool 110) may perform a planarization or polishing process such as CMP. A CMP process may include depositing a slurry (or polishing compound) onto a polishing pad. A carrier substrate including the image sensor 402 may be mounted to a carrier, which may rotate the carrier substrate as the carrier substrate is pressed against the polishing pad. The slurry and polishing pad act as an abrasive that polishes or planarizes the BSI oxide layer 320 as the carrier substrate is rotated. The polishing pad may also be rotated to ensure a continuous supply of slurry is applied to the polishing pad.

As shown in FIG. 4J, an opening 414 (or trench) may be formed in the bonding pad region 404. In particular, the opening 414 may be formed through the BSI oxide layer 320, through the metal layer 314, through the oxide layer 312, through the ARC 310, and through the substrate 304 to the STI structure 412. The opening 414 may be formed by coating the BSI oxide layer 320 with a photoresist (e.g., using the deposition tool 102), forming a pattern in the photoresist by exposing the photoresist to a radiation source (e.g., using the exposure tool 104), removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool 106), and etching the opening 414 (e.g., using the etch tool 108) based on the pattern in the photoresist.

As shown in FIG. 4K, the buffer oxide layer 322 may be formed over the BSI oxide layer 320 and over the STI structure 412 in the opening 414. In particular, a semiconductor processing tool (e.g., the deposition tool 102) may deposit the buffer oxide layer 322 using various PVD techniques, CVD techniques, and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD.

As shown in FIG. 4L, openings 416 (or vias) may be formed in the opening 414 of the bonding pad region 404. In particular, the openings 416 may be formed through the buffer oxide layer 322, through the STI structure 412, through the ILD layer 406, and to the bonding pad 410 in the IMD layer 408. The openings 416 may be formed by coating the buffer oxide layer 322 with a photoresist (e.g., using the deposition tool 102), forming a pattern in the photoresist by exposing the photoresist to a radiation source (e.g., using the exposure tool 104), removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool 106), and etching the openings 416 (e.g., using the etch tool 108) based on the pattern in the photoresist.

As shown in FIG. 4M, a bonding pad 418 may be formed in the openings 414. For example, a semiconductor processing tool (e.g., the deposition tool 102 or the plating tool 112) may form a metal layer (e.g., a copper layer, a cobalt layer, a tungsten layer, a ruthenium layer, a metal alloy layer, or another type of metal layer) on the buffer oxide layer 322, on the STI structure 412, and in the openings 416. Portions of the metal layer may be removed by coating the metal layer with a photoresist (e.g., using the deposition tool 102), forming a pattern in the photoresist by exposing the photoresist to a radiation source (e.g., using the exposure tool

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104), removing either the exposed portions or the non-exposed portions of the photoresist (e.g., using developer tool 106), and etching the portions (e.g., using the etch tool 108) based on the pattern in the photoresist to form the bonding pad 418.

As shown in FIG. 4N, the filter layer 324 may be formed for the pixel sensors 302 in the pixel array 300. The filter layer 324 may be formed by forming respective filter regions 324a-324d for the pixel sensors 302a-302d over and/or on the buffer oxide layer 322. In some implementations, a semiconductor processing tool (e.g., the deposition tool 102) may deposit the filter layer 324 using various PVD techniques, CVD techniques, and/or ALD techniques, such as sputtering, PECVD, HDP-CVD, SACVD, or PEALD. As shown in FIG. 4O, a micro-lens layer 326 including a plurality of micro-lenses is formed over and/or on the filter layer 324. The micro-lens layer 326 may include a respective micro-lens for each of the pixel sensors 302 included in the pixel array 300.

As indicated above, FIGS. 4A-4O are provided as an example. Other examples may differ from what is described with regard to FIGS. 4A-4O.

FIGS. 5A-5C are diagrams of example pixel sensor configurations described herein. The example pixel sensor configurations may include example pixel sensor configurations of visible light pixel sensors and dark pixel sensors for the pixel array 200 and/or 300 described herein. In each of the example pixel sensor configurations, each visible light pixel sensor is no more than one (1) pixel sensor away from a dark pixel sensor. In other words, each visible light pixel sensor in the example pixel sensor configurations borders at least one dark pixel sensor. In some cases, this may include a one-to-one association of visible light pixel sensors to dark pixel sensors or a one-to-many association of visible light pixel sensors to dark pixel sensors. Accordingly, the example pixel sensor configurations described in connection with FIGS. 5A-5C may provide highly accurate dark current measurements for the visible light pixel sensors, which may increase the accuracy of the dark current calibration of the visible light pixel sensors to achieve very low post-calibration dark current for the visible light pixel sensors.

As shown in FIG. 5A, an example pixel sensor configuration 500 may include a plurality of adjacent columns of visible light pixel sensors and dark pixel sensors. For example, the example pixel sensor configuration 500 may include a first column 502a of visible light pixel sensors, a first column 504a of dark pixel sensors adjacent to the first column 502a of visible light pixel sensors, a second column 502b of visible light pixel sensors adjacent to the first column 504a of dark pixel sensors, a second column 504b of dark pixel sensors adjacent to the second column 502b of visible light pixel sensors, a third column 502c of visible light pixel sensors adjacent to the second column 504b of dark pixel sensors, a third column 504c of dark pixel sensors adjacent to the third column 502c of visible light pixel sensors, and so on.

Each dark pixel sensor in a column of dark pixel sensors may generate dark current calibration information for a single (different) visible light pixel sensor in an adjacent column of visible light pixel sensors. For example, a first dark pixel sensor in the first column 504a may generate dark current calibration information for a single first visible light pixel sensor in the first column 502a, a second dark pixel sensor in the first column 504a may generate dark current calibration information for a single second visible light pixel sensor in the first column 502a (e.g., different from the single first visible light pixel sensor), and so on. A first dark

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pixel sensor in the second column **504b** may generate dark current calibration information for a single first visible light pixel sensor in the second column **502b**, a second dark pixel sensor in the second column **504b** may generate dark current calibration information for a single second visible light pixel sensor in the second column **502b** (e.g., different from the single first visible light pixel sensor), and so on. A first dark pixel sensor in the third column **504c** may generate dark current calibration information for a single first visible light pixel sensor in the third column **502c**, a second dark pixel sensor in the third column **504c** may generate dark current calibration information for a single second visible light pixel sensor in the third column **502c** (e.g., different from the single first visible light pixel sensor), and so on.

As shown in FIG. 5B, an example pixel sensor configuration **510** may include a plurality of interspersed visible light pixel sensors **512** and dark pixel sensors **514**. In particular, the visible light pixel sensors **512** and the dark pixel sensors **514** may alternate throughout the example pixel sensor configuration **510** in a checkered pattern. Each of the dark pixel sensors **514** may generate dark current calibration information for an adjacent/bordering visible light pixel sensor **512**.

As shown in FIG. 5C, an example pixel sensor configuration **520** may include a plurality of non-contiguous groups **522a-522d** of visible light pixel sensors that are surrounded by a border **524** of a plurality of dark pixel sensors. Unlike a BLC area surrounding a pixel array, which may generate a dark current measurement for the entire pixel array, each of the dark pixel sensors in the border **524** of dark pixel sensors may generate dark current calibration information for a single visible light pixel sensor in one of the groups **522a-522d**. This may increase the accuracy of the dark current calibration information and may increase the dark current performance of the visible light pixel sensors included in the example pixel sensor configuration **520**. In some implementations, a plurality of dark pixel sensors in the border **524** may generate dark current calibration information for the same visible light pixel sensor, which may further increase the accuracy of the dark current calibration information and the dark current performance of the visible light pixel sensor.

As indicated above, FIGS. 5A-5C are provided as examples. Other examples may differ from what is described with regard to FIGS. 5A-5C.

FIGS. 6A-6C are diagrams of example pixel sensor configurations described herein. The example pixel sensor configurations may include example pixel sensor configurations of visible light pixel sensors and dark pixel sensors for the pixel array **200** and/or **300** described herein. In each of the example pixel sensor configurations, one or more of the visible light pixel sensors is greater than one (1) pixel sensor away from a dark pixel sensor such that one or more dark pixel sensors are configured to generate dark current calibration information for a respective subset of visible light pixel sensors. In this way, fewer dark pixel sensors are included such that a greater quantity of visible light pixel sensors may be included. This may increase the color sensitivity and color saturation in the example pixel sensor configurations while still providing increased dark current performance. In some implementations, each visible light pixel sensor in the example pixel sensor configurations is within a range of two (2) pixel sensors away from a dark pixel sensor (e.g., to permit a greater quantity of visible light pixel sensors to be included) to approximately 100 pixel

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sensors away from a dark pixel sensor (e.g., to provide sufficient dark current calibration accuracy and sufficient dark current performance).

As shown in FIG. 6A, an example pixel sensor configuration **600** may include a plurality of visible light pixel sensors **602** and plurality of non-contiguous groups **604a-604d** of dark pixel sensors dispersed throughout the visible light pixel sensors **602**. In some implementations, the example pixel sensor configuration **600** may include at least four (4) groups of dark pixels to provide sufficient dark current calibration accuracy and sufficient dark current performance for the visible light pixel sensors **602**.

As shown in FIG. 6B, an example pixel sensor configuration **610** may include a plurality of visible light pixel sensors **612** and a plurality of dark pixel sensors **614** formed in diagonal lines through the example pixel sensor configuration **610**. In some implementations, each segment of the diagonal lines of the dark pixel sensors **614** may be configured to generate dark current calibration information for one or more quadrants (or portions of one or more quadrants) of the visible light pixel sensors **612** in the example pixel sensor configuration **610**.

As shown in FIG. 6C, an example pixel sensor configuration **620** may include a plurality of visible light pixel sensors **622** and plurality of dark pixel sensors **624** formed in a plurality of intersecting lines through the example pixel sensor configuration **620**. In some implementations, each segment of the intersecting lines of the dark pixel sensors **624** may be configured to generate dark current calibration information for one or more groups (or portions of one or more groups) of the visible light pixel sensors **622** in the example pixel sensor configuration **620**.

As indicated above, FIGS. 6A-6C are provided as examples. Other examples may differ from what is described with regard to FIGS. 6A-6C.

FIGS. 7A-7D are diagrams of an example implementation **700** described herein. The example implementation **700** may include an example of calibrating visible light information generated by a plurality of visible light pixel sensors included in the pixel array **300** based on dark current calibration information generated by a plurality of dark pixel sensors included in the pixel array **300**.

As shown in FIG. 7A, the pixel array **300** may be included in the image sensor **402**, which may be included in a device **702**. The device **702** may further include image processor **704**, communicatively coupled with the image sensor **402**, configured to process visible light information generated by the pixel array **300** to generate images and/or video. Moreover, the image processor **704** may be configured to process dark current calibration information generated by the pixel array **300** to calibrate the visible light information to increase the dark current performance of the pixel array **300**. The device **702** may include a semiconductor die, a system on chip (SoC), a semiconductor package, a camera, a smartphone, or another type of device. In some implementations, the image processor **704** is integrated into the image sensor **402** or vice-versa. In some implementations, the image sensor **402** and the image processor **704** are included in separate devices. For example, the image processor **704** may be included in a computer or another device remote from the image sensor **402**.

As shown in FIG. 7B, the plurality of visible light pixel sensors of the pixel array **300** may be configured to generate (and may generate) visible light information **706**. The visible light information **706** may include respective photocurrents for each of the plurality of visible light pixel sensors. The magnitude of the respective photocurrents may represent or

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correspond to the intensity of various colors or wavelengths of incident light captured by the plurality of visible light pixel sensors. Thus, the visible light information 706 may include light intensity information across a broad range of color components of the incident light.

As further shown in FIG. 7B, various levels of dark current (in electrons per second (e-/sec)) may be present in (or may affect) the photocurrent levels included in the visible light information 706 for each of the plurality of visible light pixel sensors. In particular, dark current may artificially increase the magnitude of a photocurrent of a visible light pixel sensor, which may result in an elevated brightness level in the pixel associated with the visible light pixel sensor (e.g., in an image or a video). This may negatively affect the black level of the visible light pixel sensor.

As further shown in FIG. 7B, the dark pixel sensors may be configured to generate (and may generate) dark current calibration information 708 for the plurality of visible light pixel sensors. In some implementations, each dark pixel sensor is configured to generate (and generates) dark current calibration information for a single visible light pixel sensor. The dark current calibration information, associated with a visible light pixel sensor, may include a dark current measurement (e.g., in electrons per second (e-/sec)) for the visible light pixel sensor. The dark current measurement for a visible light pixel sensor may represent an estimation of the level of dark current in the visible light information generated by the visible light pixel sensor. Accordingly, the dark pixel sensors may be configured to generate (and may generate) the dark current calibration information approximately simultaneously with the visible light pixel sensors generating the visible light information, 1 or 2 milliseconds prior to visible light pixel sensors generating the visible light information, and/or 1 or 2 milliseconds after the visible light pixel sensors generate the visible light information.

As shown in FIG. 7C, the image sensor 402 may be configured to provide (and may provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704. In some implementations, the image sensor 402 may be configured to provide (and may provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704 on one or more metallization layers of a die or an SoC. In some implementations, the image sensor 402 may be configured to provide (and may provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704 over one or more circuit traces of an integrated circuit or a printed circuit board (PCB). In some implementations, the image sensor 402 may be configured to provide (and may provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704 over a communication cable such as an Ethernet cable or a universal serial bus (USB) cable. In some implementations, the image sensor 402 may be configured to provide (and may provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704 over a public network and/or a private network, such as a wireless network, a wired network, a telecommunications network, a peer-to-peer network, a local area network (LAN), and/or the Internet.

In some implementations, the image sensor 402 may be configured to automatically provide (and may automatically provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704 based on the visible light information 706 and the dark current calibration information 708 being generated. In some implementations, the image sensor 402 may be con-

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figured to provide (and may provide) the visible light information 706 and the dark current calibration information 708 to the image processor 704 based on a request from the image processor 704.

As shown in FIG. 7D, the image processor 704 may be configured to generate (and may generate) calibrated visible light information 710 from the visible light information 706 based on the dark current calibration information 708, which may be referred to as dark current calibration. To generate the calibrated visible light information 710 for a visible light pixel sensor, the image processor 704 may be configured to remove, subtract, and/or adjust the photocurrent generated by the visible light pixel sensor (e.g., included in the visible light information 706 associated with the visible light pixel sensor) based on the dark current measurement associated with the visible light pixel sensor (e.g., included in the dark current calibration information 708 associated with the visible light pixel sensor). The image processor 704 may be configured to remove, subtract, and/or adjust the photocurrent generated by each visible light pixel sensor based on the respective dark current measurements associated with each of the visible light pixel sensors such that each of the visible light pixel sensors is individually calibrated. As shown in FIG. 7D, the resulting dark current (e.g., in electrons per second (e-/sec)) for each of the visible light pixel sensors after dark current calibration is approximately zero or much lower than prior to dark current calibration. The individualized calibration of each of the visible light pixel sensors may be capable of addressing relatively large differences in dark current in the visible light information 706 that might not otherwise be correctable by an average dark current measurement of a BLC area.

In some implementations, the image processor 704 is configured to generate (and generates) the calibrated visible light information 710 in real-time or near-real-time with the generation of the visible light information 706. In these cases, the image processor 704 may generate a dark-current-calibrated stream of images and/or a video based on the generated dark current calibration information 708. In some implementations, the image processor 704 is configured to generate (and generates) the calibrated visible light information 710 after a longer time duration relative to the generation of the visible light information 706. In these examples, the image processor 704 may be configured to store (and may store) visible light information 706 and the dark current calibration information 708 in a data store such as a cache, a volatile or non-volatile memory, a storage device, a file system, and/or another type of data structure. The data store may be included in the image processor 704, may be included in the device 702, or may be remote from the device 702 such as a cloud storage, a data center, a network attached storage (NAS), or a computing device. The image processor 704 may be configured to obtain (and may obtain) the visible light information 706 and the dark current calibration information 708 from the data store and may generate the calibrated visible light information 710 based on obtaining the visible light information 706 and the dark current calibration information 708.

In some implementations, the image processor 704 is configured to automatically generate (and automatically generates) the calibrated visible light information 710 based on receiving the visible light information 706 and the dark current calibration information 708. In some implementations, the image processor 704 is configured to generate (and generates) the calibrated visible light information 710 as a part of a process to convert raw visible light information 706 to an image file format (e.g., JPEG, tag image file format



(TIFF), portable network graphics (PNG)) or a video file format (e.g., MPEG, audio video interleave (AVI)). In some implementations, the image processor 704 is configured to generate (and generates) the calibrated visible light information 710 based on input to the device 702.

As indicated above, FIGS. 7A-7D are provided as an example. Other examples may differ from what is described with regard to FIGS. 7A-7D.

FIG. 8 is a diagram of example components of a device 800, which may correspond to the image sensor 402, the device 702, and/or the image processor 704. In some implementations, the image sensor 402, the device 702, and/or the image processor 704 may include one or more devices 800 and/or one or more components of device 800. As shown in FIG. 8, device 800 may include a bus 810, a processor 820, a memory 830, a storage component 840, an input component 850, an output component 860, and a communication component 870.

Bus 810 includes a component that enables wired and/or wireless communication among the components of device 800. Processor 820 includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor 820 is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor 820 includes one or more processors capable of being programmed to perform a function. Memory 830 includes a random access memory, a read only memory, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory).

Storage component 840 stores information and/or software related to the operation of device 800. For example, storage component 840 may include a hard disk drive, a magnetic disk drive, an optical disk drive, a solid state disk drive, a compact disc, a digital versatile disc, and/or another type of non-transitory computer-readable medium. Input component 850 enables device 800 to receive input, such as user input and/or sensed inputs. For example, input component 850 may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system component, an accelerometer, a gyroscope, and/or an actuator. Output component 860 enables device 800 to provide output, such as via a display, a speaker, and/or one or more light-emitting diodes. Communication component 870 enables device 800 to communicate with other devices, such as via a wired connection and/or a wireless connection. For example, communication component 870 may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

Device 800 may perform one or more processes described herein. For example, a non-transitory computer-readable medium (e.g., memory 830 and/or storage component 840) may store a set of instructions (e.g., one or more instructions, code, software code, and/or program code) for execution by processor 820. Processor 820 may execute the set of instructions to perform one or more processes described herein. In some implementations, execution of the set of instructions, by one or more processors 820, causes the one or more processors 820 and/or the device 800 to perform one or more processes described herein. In some implementations, hardware circuitry may be used instead of or in combination with the instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

The number and arrangement of components shown in FIG. 8 are provided as an example. Device 800 may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. 8. Additionally, or alternatively, a set of components (e.g., one or more components) of device 800 may perform one or more functions described as being performed by another set of components of device 800.

FIG. 9 is a flowchart of an example process 900 associated with forming a pixel array. In some implementations, one or more process blocks of FIG. 9 may be performed by one or more semiconductor processing tools (e.g., one or more of the semiconductor processing tools 102-114). Additionally, or alternatively, one or more process blocks of FIG. 9 may be performed by one or more components of device 800, such as processor 820, memory 830, storage component 840, input component 850, output component 860, and/or communication component 870.

As shown in FIG. 9, process 900 may include forming a plurality of photodiodes in a substrate of an image sensor (block 910). For example, the one or more semiconductor processing tools 102-114 may form a plurality of photodiodes 306 in the substrate 304 of the image sensor 402, as described above.

As further shown in FIG. 9, process 900 may include forming a metal layer over the plurality of photodiodes in the substrate (block 920). For example, the one or more semiconductor processing tools 102-114 may form the metal layer 314 over the plurality of photodiodes 306 in the substrate 304, as described above.

As further shown in FIG. 9, process 900 may include removing portions of the metal layer to form openings over a first subset of the plurality of photodiodes, where the openings are included for a plurality of visible light pixel sensors of the image sensor, and where remaining portions of the metal layer include light blocking elements over a second subset of the plurality of photodiodes for a plurality of dark pixel sensors of the image sensor (block 930). For example, the one or more semiconductor processing tools 102-114 may remove portions of the metal layer to form the openings 316 over a first subset of the plurality of photodiodes 306, as described above. In some implementations, the openings 316 are included for a plurality of visible light pixel sensors (e.g., pixel sensors 202, 302a, 302c, 512, 602, 612, and/or 622) of the image sensor 402. In some implementations, remaining portions of the metal layer 314 include the light blocking elements 318 over a second subset of the plurality of photodiodes 306 for a plurality of dark pixel sensors (e.g., pixel sensors 202, 302b, 302d, 514, 614, and/or 624) of the image sensor 402.

As further shown in FIG. 9, process 900 may include filling the openings with an oxide layer (block 940). For example, the one or more semiconductor processing tools 102-114 may fill the openings 316 with the oxide layer 320, as described above.

Process 900 may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

In a first implementation, a first subset of the plurality of visible light pixel sensors is included in a first column 502a of a pixel array (e.g., pixel array 200 and/or 300) of the image sensor, a first subset of the plurality of dark pixel sensors is included in a second column 504a of the pixel array adjacent to the first column, a second subset of the plurality of visible light pixel sensors is included in a third column 502b of the pixel array adjacent to the second



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column, and a second subset of the plurality of dark pixel sensors is included in a fourth column **504b** of the pixel array adjacent to the third column. In a second implementation, alone or in combination with the first implementation, the plurality of visible light pixel sensors and the plurality of dark pixel sensors are interspersed in a pixel array (e.g., pixel array **200** and/or **300**) of the image sensor.

In a third implementation, alone or in combination with one or more of the first and second implementations, non-contiguous groups of the plurality of visible light pixel sensors (e.g., groups **522a**, **522b**, **522c**, and/or **522c**) are included in a pixel array (e.g., pixel array **200** and/or **300**) of the image sensor, and the non-contiguous groups of the plurality of visible light pixel sensors are at least partially surrounded by the plurality of dark pixel sensors in the pixel array. In a fourth implementation, alone or in combination with one or more of the first through third implementations, the metal layer includes a plurality of layers (e.g., layers **328**, **330**, and/or **332**), where a respective thickness of each of the plurality of layers is equal to or less than approximately 5000 angstroms.

Although FIG. **9** shows example blocks of process **900**, in some implementations, process **900** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **9**. Additionally, or alternatively, two or more of the blocks of process **900** may be performed in parallel.

FIG. **10** is a flowchart of an example process **1000** associated with generating calibrated visible light information. In some implementations, one or more process blocks of FIG. **10** may be performed by a device (e.g., the device **702**). In some implementations, one or more process blocks of FIG. **10** may be performed by another device or a group of devices separate from or including the device, such as a pixel array (e.g., pixel array **200** and/or **300**), an image sensor (e.g., the image sensor **402**), and/or an image processor (e.g., the image processor **704**), among other examples. Additionally, or alternatively, one or more process blocks of FIG. **10** may be performed by one or more components of device **800**, such as processor **820**, memory **830**, storage component **840**, input component **850**, output component **860**, and/or communication component **870**.

As shown in FIG. **10**, process **1000** may include generating visible light information based on incident light into a plurality of visible light pixel sensors (block **1010**). For example, the device (e.g., using a plurality of visible light pixel sensors, such as the pixel sensors **202**, **302a**, **302c**, **512**, **602**, **612**, and/or **622**) may generate the visible light information **706** based on incident light into a plurality of visible light pixel sensors, as described above.

As further shown in FIG. **10**, process **1000** may include generating dark current calibration information for the plurality of visible light pixel sensors (block **1020**). For example, the device (e.g., using a plurality dark pixel sensors, such as pixel sensors **202**, **302b**, **302d**, **514**, **614**, and/or **624**) may generate dark current calibration information for the plurality of visible light pixel sensors, as described above.

As further shown in FIG. **10**, process **1000** may include providing the visible light information and the dark current calibration information to an image processor (block **1030**). For example, the device may (e.g., using the image sensor **402**) may provide the visible light information **706** and the dark current calibration information **708** to the image processor **704**, as described above.

As further shown in FIG. **10**, process **1000** may include generating, based on the dark current calibration informa-

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tion, calibrated visible light information from the visible light information (block **1040**). For example, the device (e.g., using the image processor **704**) may generate, based on the dark current calibration information **708**, calibrated visible light information **710** from the visible light information **706**, as described above.

Process **1000** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

In a first implementation, generating the calibrated visible light information includes generating the calibrated visible light information in near-real-time as the visible light information is generated. In a second implementation, alone or in combination with the first implementation, generating the calibrated visible light information includes storing the visible light information and the dark current calibration information in a data store, obtaining the visible light information and the dark current calibration information from the data store after storing the visible light information and the dark current calibration information in the data store, and generating the calibrated visible light information after obtaining the visible light information and the dark current calibration information from the data store.

In a third implementation, alone or in combination with one or more of the first and second implementations, generating the calibrated visible light information includes generating respective calibrated visible light information for each of the plurality of visible light pixel sensors. In a fourth implementation, alone or in combination with one or more of the first through third implementations, generating the dark current calibration information includes generating a dark current measurement for a single visible light pixel sensor of the plurality of visible light pixel sensors, and generating the calibrated visible light information includes subtracting the dark current measurement from a visible light current measurement associated with the single visible light pixel sensor.

In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, generating the dark current calibration information includes generating another dark current measurement for another single visible light pixel sensor of the plurality of visible light pixel sensors, the dark current measurement and the other dark current measurement being different dark current measurements, and generating the calibrated visible light information includes subtracting the other dark current measurement from another visible light current measurement associated with the other single visible light pixel sensor.

Although FIG. **10** shows example blocks of process **1000**, in some implementations, process **1000** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **10**. Additionally, or alternatively, two or more of the blocks of process **1000** may be performed in parallel.

In this way, a pixel array includes a plurality of dark pixel sensors configured to generate dark current calibration information for a plurality of visible light pixel sensors included in the pixel array. In particular, the plurality of dark pixel sensors may generate respective dark current measurements for each of the plurality of visible light pixel sensors (e.g., one-to-one or per-pixel-sensor dark current measurements) or for small subsets of the plurality of visible light pixel sensors. In this way, each of the plurality of visible light pixel sensors may be individually calibrated (or small subsets of the plurality of visible light pixel sensors may be individually calibrated) based on an estimated dark current

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experienced by each of the plurality of visible light pixel sensors. This individual dark current calibration may enable more accurate dark current calibration of the visible light pixel sensors included in the pixel array, and may be used to account for large differences in estimated dark currents for the visible light pixel sensors.

As described in greater detail above, some implementations described herein provide a pixel array. The pixel array includes a plurality of visible light pixel sensors configured to generate visible light information. The pixel array includes a plurality of dark pixel sensors configured to generate dark current calibration information. A subset of the dark current calibration information generated by a dark pixel sensor of the plurality of dark pixel sensors is to be used for calibrating a subset of the visible light information generated by one or more of the plurality of visible light pixel sensors.

As described in greater detail above, some implementations described herein provide a method. The method includes forming a plurality of photodiodes in a substrate of an image sensor. The method includes forming a metal layer over the plurality of photodiodes in the substrate. The method includes removing portions of the metal layer to form openings over a first subset of the plurality of photodiodes, where the openings are included for a plurality of visible light pixel sensors of the image sensor, and where remaining portions of the metal layer include light blocking elements over a second subset of the plurality of photodiodes for a plurality of dark pixel sensors of the image sensor. The method includes filling the openings with an oxide layer.

As described in greater detail above, some implementations described herein provide a device. The device includes a plurality of visible light pixel sensors configured to generate visible light information based on incident light into the plurality of visible light pixel sensors. The device includes a plurality of dark pixel sensors configured to generate dark current calibration information for the plurality of visible light pixel sensors. The device includes an image sensor configured to provide the visible light information and the dark current calibration information to an image processor. The device includes the image processor configured generate, based on the dark current calibration information, calibrated visible light information from the visible light information.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A pixel array, comprising:

- a plurality of visible light pixel sensors configured to generate visible light information; and
- a plurality of dark pixel sensors configured to generate dark current calibration information,

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wherein each visible light pixel sensor, of the plurality of visible light pixel sensors, is within two pixel sensors away from a dark pixel sensor of the plurality of dark pixel sensors,

wherein at least one visible light pixel sensor, of the plurality of visible light pixel sensors, is greater than one pixel sensor away from each dark pixel sensor of the plurality of dark pixel sensors,

wherein the plurality of dark pixel sensors comprise a two-dimensional array of non-contiguous groups of dark pixel sensors, and

wherein each of the non-contiguous groups of dark pixel sensors are surrounded by a border of visible light pixel sensors of the plurality of visible light pixel sensors.

2. The pixel array of claim 1, wherein the two-dimensional array of non-contiguous groups of dark pixel sensors includes at least four groups of dark pixel sensors.

3. The pixel array of claim 1, wherein each of the non-contiguous groups of dark pixel sensors are two-dimensional arrays.

4. The pixel array of claim 1, wherein the plurality of dark pixel sensors are configured to generate respective dark current measurements for each of the plurality of visible light pixel sensors.

5. The pixel array of claim 1, wherein a quantity of the plurality of dark pixel sensors is configured to satisfy at least one of:

- a dark current performance parameter,
- a color saturation performance parameter,
- a color sensitivity performance parameter, or
- one or more other types of performance parameters.

6. The pixel array of claim 1, further comprising: a light blocking element over a photodiode of a dark pixel sensor of the plurality of dark pixel sensors.

7. The pixel array of claim 6, wherein the light blocking element fully extends between openings in a metal layer above a subset of the plurality of visible light pixel sensors.

8. A method, comprising:

forming a plurality of visible light pixel sensors configured to generate visible light information; and

forming a plurality of dark pixel sensors configured to generate dark current calibration information,

wherein each visible light pixel sensor, of the plurality of visible light pixel sensors, is within two pixel sensors away from a dark pixel sensor of the plurality of dark pixel sensors,

wherein at least one visible light pixel sensor, of the plurality of visible light pixel sensors, is greater than one pixel sensor away from each dark pixel sensor of the plurality of dark pixel sensors,

wherein the plurality of dark pixel sensors are formed as a two-dimensional array of non-contiguous groups of dark pixel sensors, and

wherein each of the non-contiguous groups of dark pixel sensors are formed to be surrounded by a border of visible light pixel sensors of the plurality of visible light pixel sensors.

9. The method of claim 8, wherein forming the plurality of visible light pixel sensors comprises:

- forming a plurality of photodiodes;
- forming a metal layer over the plurality of photodiodes; and

removing portions of the metal layer to form openings over a subset of the plurality of photodiodes.

10. The method of claim 9, wherein forming the plurality of visible light pixel sensors further comprises: filling the openings with an oxide layer.

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11. The method of claim 9, wherein remaining portions of the metal layer, based on removing the portions of the metal layer, are associated with the plurality of dark pixel sensors.

12. The method of claim 8, wherein forming the plurality of dark pixel sensors comprises:

forming at least four non-contiguous groups of dark pixel sensors.

13. The method of claim 12, wherein each of the at least four non-contiguous groups of dark pixel sensors are two dimensional arrays.

14. The method of claim 8, further comprises:

forming a respective micro-lens over each of the plurality of visible light pixel sensors and the plurality of dark pixel sensors.

15. A device, comprising:

a plurality of visible light pixel sensors configured to generate visible light information;

a plurality of dark pixel sensors configured to generate dark current calibration information,

wherein each visible light pixel sensor, of the plurality of visible light pixel sensors, is within two pixel sensors away from a dark pixel sensor of the plurality of dark pixel sensors,

wherein at least one visible light pixel sensor, of the plurality of visible light pixel sensors, is greater than one pixel sensor away from each dark pixel sensor of the plurality of dark pixel sensors,

wherein the plurality of dark pixel sensors comprise a two-dimensional array of non-contiguous groups of dark pixel sensors, and

wherein each of the non-contiguous groups of dark pixel sensors are surrounded by a border of visible light pixel sensors of the plurality of visible light pixel sensors;

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an image sensor configured to output the visible light information and the dark current calibration information; and

an image processor configured to:

receive the visible light information and the dark current calibration information, and

generate calibrated visible light information based on the dark current calibration information and the visible light information.

16. The device of claim 15, wherein the image processor, to generate the calibrated visible light information, is configured to:

generate the calibrated visible light information in real-time or near-real-time as the visible light information is generated.

17. The device of claim 15, wherein the image processor, to generate the calibrated visible light information, is configured to:

at least one of remove, subtract, or adjust photocurrent generated by the plurality of visible light pixel sensors.

18. The device of claim 15, wherein the image processor, to generate the calibrated visible light information, is configured to:

at least one of remove, subtract, or adjust photocurrent generated by each of the plurality of visible light pixel sensors based on respective dark current measurements associated with each of the plurality of visible light pixel sensors.

19. The device of claim 15, wherein the image sensor is configured to provide the visible light information and the dark current calibration information to the image processor.

20. The device of claim 15, wherein the visible light information includes respective photocurrents for each of the plurality of visible light pixel sensors.

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